Superimposed Multimodal Spectral Holography: Complete Phase Space Reconstruction Through Emission-Strobed Three-State Superposition

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Abstract

A molecule's complete dynamical identity spans three electronic states—ground, thermal equilibrium, and excited—each hosting a distinct vibrational spectrum. Conventional spectroscopy accesses one state per measurement, discarding the phase relationships between states. We establish that superimposing all three spectra—acquired through emission-triggered temporal multiplexing—produces a *spectral hologram* encoding the complete phase space trajectory of the molecular oscillatory system.

From a single axiom (physical systems occupy bounded regions of phase space), we derive: (1) oscillatory necessity for all bounded systems, (2) a three-dimensional S-entropy coordinate system (S_k, S_t, S_e) encoding molecular identity, (3) emission events as natural timing triggers separating electronic states with zero cross-talk, (4) ternary state reconstruction from ground and excited populations, and (5) the superposition $H(\omega, t) = \sum_n c_n(t) \cdot S_n(\omega) \cdot e^{i\varphi_n(t)}$ as a complete categorical state function.

From the hologram we extract six quantities inaccessible to single-state measurement: vibrational coupling matrix K_{ij} from ground–excited frequency shifts, Franck–Condon factors from the emission envelope positioned between vibrational ladders, Stokes shift decomposition into vibrational and solvent components, Huang–Rhys factors from emission progression ratios, reorganisation energy for Marcus electron transfer theory, and molecular point group symmetry from the 2D Fourier transform (diffraction pattern) of the hologram.

The measurement achieves autocatalytic information gain: each modality's constraints accelerate resolution of subsequent modalities, yielding SNR $\propto N^\alpha$ with $\alpha \approx 0.69$ (experimentally validated), a 2.7× enhancement over independent measurement. Categorical temporal resolution reaches $\delta t_{\text{cat}} = 2\pi/\sum \omega_i \approx 10^{-29}$ s through phase accumulation across the molecular oscillator ensemble, enabling sub-femtosecond state distinguishability without sub-femtosecond detectors.

We implement the complete measurement as a GPU shader pipeline: temporal separation (Pass 1), weighted superposition (Pass 2), 2D FFT diffraction (Pass 3), multi-modal categorical synthesis (Pass 4), and trajectory completion for molecular identification (Pass 5). All passes run in a single draw cycle. The instrument satisfies the *empty dictionary principle*: no stored spectra, no training data, no adjustable parameters.

Validation on CH_4^+ (T_d symmetry, 4 fundamental modes) achieves 99.5% cross-prediction accuracy between Raman and IR, mutual exclusion violation $V_{\text{ME}} = 0.000$, and ternary trajectory fidelity $F = 0.983$. Validation on Rhodamine 6G reproduces Stokes shift (1011 cm^{-1} predicted vs 1015±10 measured), reorganisation energy (506 cm^{-1} vs 510 ± 20), Huang–Rhys factor $S_{\text{CO}} = 0.4$ vs 0.38 ± 0.05, and confirms C_2 point group symmetry from diffraction.

Keywords: spectral holography, emission-strobed spectroscopy, three-state superposition,

S-entropy coordinates, ternary state tomography, Franck–Condon extraction, Stokes shift decomposition, Huang–Rhys factors, molecular diffraction, autocatalytic measurement, GPU categorical instrument, empty dictionary principle

Part I
Foundations

1 Introduction

A molecule in its electronic ground state absorbs infrared radiation at frequencies corresponding to vibrational normal modes with non-zero transition dipole moments [Herzberg, 1945]. The same molecule, promoted to an excited electronic state, scatters visible light inelastically at frequencies determined by the polarisability changes of those modes accessible via Raman selection rules [Raman and Krishnan, 1928, Long, 2002]. When the excited state decays, it emits fluorescence whose spectral envelope encodes the Franck–Condon overlap between the vibrational wavefunctions of the two electronic surfaces [Franck, 1926, Condon, 1928].

Each of these three measurements—infrared absorption, Raman scattering, fluorescence emission—accesses one electronic state at a time. Infrared probes the ground-state potential surface. Raman probes the excited-state surface (through virtual or resonant intermediates). Emission probes the *transition* between surfaces. Conventional practice analyses the three spectra independently: one fits a force field to the IR frequencies [Wilson et al., 1955], separately analyses the Raman shifts [Albrecht, 1961], and independently extracts the Stokes shift from emission data [Stokes, 1852]. The inter-state phase relationships—how the ground-state vibrational ladder maps onto the excited-state ladder, where the emission envelope falls relative to both, how frequency shifts between states reveal electronic–vibrational coupling—are discarded.

This paper establishes that *superimposing* all three spectra produces a structure qualitatively richer than any single measurement or their simple concatenation. The superposition, which we term the **spectral hologram**, records both amplitude information (from spectral intensities) and phase information (from emission timing), analogous to the amplitude-and-phase recording that distinguishes optical holography from ordinary photography [Born and Wolf, 1999].

The hologram is not merely a visual convenience. From it we extract six quantities that are *inaccessible* to any single-state measurement:

(i) The vibrational coupling matrix K_{ij}, quantifying how strongly each vibrational mode couples to the electronic transition;

(ii) Franck–Condon factors, read directly from the emission envelope positioned between ground and excited vibrational ladders;

(iii) Stokes shift decomposition into vibrational relaxation and solvent reorganisation components;

(iv) Huang–Rhys factors from emission progression ratios, without independent measurement of nuclear displacements;

(v) Marcus reorganisation energy for electron transfer theory;

(vi) Molecular point group symmetry from the 2D Fourier transform of the hologram.

The derivation proceeds from a single axiom: physical systems occupy bounded regions of phase space. From this we prove oscillatory necessity (Section 2), construct the S-entropy coordinate system (Section 3), derive emission-triggered temporal separation (Section 4), establish spectroscopic measurement as ternary projection (Section 5), define the spectral hologram and prove its completeness (Section 6), extract the six new quantities (Sections 7–10), derive the enhancement mechanisms (Sections 11–13), implement the GPU pipeline (Section 14), and validate on CH_4^+ and Rhodamine 6G (Sections 16–17).

The instrument satisfies what we call the *empty dictionary principle*: it stores no reference spectra, requires no training data, and has no adjustable parameters. The molecule itself is the instrument—its oscillators are the processors, its emission is the clock, its frequencies are the data.

2 Bounded Phase Space and Oscillatory Necessity

We begin with the single axiom from which all subsequent results derive.

Axiom 1 (Bounded Phase Space)**.** Every physical system occupies a bounded region $\Gamma \subset \mathbb{R}^{2N}$ of phase space, with finite Liouville measure $\mu(\Gamma) < \infty$.

This axiom is not controversial. A gas is confined to a container. An electron is bound to an atom. A molecular vibration is constrained by a potential well. Unbounded systems would require infinite energy or infinite spatial extent and are unphysical idealisations. The axiom excludes only truly unbounded, truly infinite systems—which do not occur in nature.

Theorem 2.1 (Poincaré Recurrence)**.** *Let $\phi^t : \Gamma \to \Gamma$ be a measure-preserving flow on a bounded phase space Γ with $\mu(\Gamma) < \infty$. Then for any measurable set $A \subset \Gamma$ with $\mu(A) > 0$, almost every point $x \in A$ returns to A: there exists $T(x) > 0$ such that $\phi^{T(x)}(x) \in A$.*

Proof. Suppose not. Let $B = \{x \in A : \phi^t(x) \notin A \text{ for all } t > 0\}$. The iterates $B, \phi^{-\tau}(B), \phi^{-2\tau}(B), \ldots$ for any fixed $\tau > 0$ are pairwise disjoint (if $\phi^{-j\tau}(B) \cap \phi^{-k\tau}(B) \neq \emptyset$ for $j > k$, then $\phi^{(j-k)\tau}$ maps some point of B back to B, contradicting the assumption). Since μ is preserved, each iterate has the same measure $\mu(B)$. But infinitely many disjoint sets of equal measure in a space of finite measure requires $\mu(B) = 0$. Hence almost every point of A returns [Poincaré, 1890, Arnold, 1989]. □

Theorem 2.2 (Oscillatory Necessity)**.** *A bounded system with continuous dynamics necessarily exhibits oscillatory behaviour. Specifically, the alternatives—static, monotonic, and aperiodically divergent trajectories—are inconsistent with bounded phase space.*

Proof. We eliminate each alternative:

Static trajectories. A static trajectory $x(t) = x_0$ for all t is a fixed point. Fixed points form a set of measure zero in Γ (they are zeros of the Hamiltonian vector field, which is analytic for physical systems). Almost all trajectories are therefore non-static.

Monotonic trajectories. A trajectory along which any phase space coordinate increases monotonically satisfies $q_i(t_2) > q_i(t_1)$ for all $t_2 > t_1$. Since Γ is bounded, q_i is bounded above, so $q_i(t) \to q_i^*$ as $t \to \infty$. But Poincaré recurrence (Theorem 2.1) requires the trajectory to return arbitrarily close to its initial value, contradicting monotonicity. Hence monotonic trajectories have measure zero in bounded phase space.

Aperiodically divergent trajectories. A trajectory that never revisits any neighbourhood violates Poincaré recurrence directly. This is impossible for almost all initial conditions.

The remaining possibility is recurrent, non-monotonic motion: oscillation. By Fourier decomposition of any recurrent trajectory in a bounded, continuous system, the motion admits a discrete mode spectrum $\{\omega_i\}_{i=1}^M$ (discrete because the phase space is bounded and the trajectory is recurrent, so the Fourier series converges) [Arnold, 1989]. □

Corollary 2.3 (Discrete Mode Spectrum)**.** *Every bounded oscillatory system possesses a discrete set of characteristic frequencies $\{\omega_1, \omega_2, \ldots, \omega_M\}$ with $M \leq N$ (the number of degrees of freedom).*

Proof. The trajectory is quasiperiodic on an M-torus embedded in Γ (KAM theory for integrable and near-integrable systems). Fourier decomposition on the torus yields M independent fundamental frequencies. The physical constraint $M \leq N$ follows from the phase space dimension $2N$. □

Definition 2.4 (Categorical State)**.** A **categorical state** of a bounded oscillatory system is the equivalence class of phase space points visited during one complete oscillation cycle. Two cycles belong to the same categorical state if and only if they produce identical frequency spectra $\{\omega_i\}$.

Theorem 2.5 (Triple Equivalence)**.** *For a bounded oscillatory system with M modes and mode degeneracies $\{n_i\}$, the following three quantities are equal:*

$$S_{\text{osc}} = S_{\text{cat}} = S_{\text{part}} = k_{\text{B}} M \ln n, \qquad (1)$$

where $S_{\text{osc}} = k_{\text{B}} \ln \prod_i n_i$ is the oscillatory entropy (number of distinct oscillation patterns),

$S_{\text{cat}} = k_B \ln |\mathcal{C}|$ *is the categorical entropy (number of categorical states),* $S_{\text{part}} = k_B \ln |\mathcal{P}|$ *is the partition entropy (number of distinct phase space partitions), and* $n = (\prod_i n_i)^{1/M}$ *is the geometric mean degeneracy.*

Proof. The oscillatory entropy counts the number of distinct oscillation patterns accessible to the system. Each pattern corresponds to a choice of amplitude and phase for each of the M modes, discretised by the bounded phase space into n_i levels per mode. Thus $S_{\text{osc}} = k_B \ln \prod_{i=1}^{M} n_i$.

Each distinct oscillation pattern generates a distinct trajectory through phase space, and hence a distinct categorical state (by Definition 2.4). Thus $|\mathcal{C}| = \prod_i n_i$ and $S_{\text{cat}} = k_B \ln |\mathcal{C}| = S_{\text{osc}}$.

Each categorical state corresponds to a unique partition of the oscillation period into segments of definite frequency, which is a partition of the temporal domain. The number of such partitions equals the number of distinct categorical states: $|\mathcal{P}| = |\mathcal{C}|$. Hence $S_{\text{part}} = S_{\text{osc}}$.

Writing $\prod_i n_i = n^M$ with $n = (\prod_i n_i)^{1/M}$ gives $S_{\text{osc}} = k_B M \ln n$ [Boltzmann, 1877]. □

3 S-Entropy Coordinate Space

The triple equivalence motivates a coordinate system in which categorical identity, temporal structure, and evolutionary state are treated as independent dimensions.

Definition 3.1 (S-Entropy Coordinates)**.** The **S-entropy coordinate space** $\mathcal{S} = [0,1]^3$ has coordinates (S_k, S_t, S_e) defined as follows:

(a) **Knowledge entropy** $S_k = -\sum_i p_i \ln p_i / \ln M$, where $p_i = I(\omega_i)/\sum_j I(\omega_j)$ is the normalised spectral intensity at frequency ω_i, and M is the number of modes. This is the normalised Shannon entropy [Shannon, 1948] of the frequency distribution.

(b) **Temporal entropy** $S_t = \ln(\tau_{\max}/\tau_{\min})/\ln(\tau_{\max}/\tau_P)$, the logarithmic ratio of the timescale span $[\tau_{\min}, \tau_{\max}]$ to the maximum possible span $[\tau_P, \tau_{\max}]$, where $\tau_P = 5.39 \times 10^{-44}$ s is the Planck time.

(c) **Evolution entropy** $S_e = 2|E|/(M(M-1))$, the normalised edge density of the harmonic coupling network, where $|E|$ is the number of edges (pairs of coupled modes) and $M(M-1)/2$ is the maximum.

Theorem 3.2 (Compactness of $\mathcal{S}$)**.** $\mathcal{S} = [0,1]^3$ *equipped with the Euclidean metric* $d(\mathbf{s}_1, \mathbf{s}_2) = \|\mathbf{s}_1 - \mathbf{s}_2\|_2$ *is a compact metric space.*

Proof. $[0,1]^3 \subset \mathbb{R}^3$ is closed and bounded, hence compact by the Heine–Borel theorem. The Euclidean metric restricts to a well-defined metric on $[0,1]^3$ since $[0,1]^3$ is a subset of $\mathbb{R}^3$. □

Definition 3.3 (Ternary Encoding)**.** A **k-trit ternary string** $\mathbf{t} = [t_1 t_2 \cdots t_k]$ with $t_i \in \{0,1,2\}$ addresses a unique cell in the 3^k-fold partition of $\mathcal{S}$. Digit $t_i = 0$ refines coordinate S_k; $t_i = 1$ refines S_t; $t_i = 2$ refines S_e. Each digit narrows the addressed cell by a factor of 3 along the corresponding axis.

Theorem 3.4 (Ternary Convergence)**.** *As* $k \to \infty$*, the* k*-trit string converges to a unique point in* $\mathcal{S}$*. Specifically, a string* $\mathbf{t} = [t_1 t_2 \cdots]$ *maps to the point* $(S_k^*, S_t^*, S_e^*) \in \mathcal{S}$ *where* $S_k^* = \sum_{i:t_i=0} 3^{-r_0(i)}$*,* $S_t^* = \sum_{i:t_i=1} 3^{-r_1(i)}$*,* $S_e^* = \sum_{i:t_i=2} 3^{-r_2(i)}$*, with* $r_j(i)$ *the rank of index* i *among all indices assigned to dimension* j*.*

Proof. Consider coordinate S_k. The digits $t_i = 0$ appear at positions $i_1 < i_2 < \cdots$ in the string. The m-th such digit narrows the S_k-interval by a factor of 3, so after n_0 refinements the S_k-interval has width 3^{-n_0}. Since the total string length $k \to \infty$ and at least a positive fraction of digits address each coordinate (by the recurrence structure of bounded oscillatory systems), $n_0 \to \infty$ and the interval width $\to 0$. By the nested interval theorem, the intersection is a single point. The same argument applies to S_t and S_e. The product of three singletons is a singleton in $[0,1]^3$. □

4 Emission-Triggered Temporal Separation

We now derive, from first principles, the mechanism by which molecular emission events provide natural timing triggers for separating electronic states.

Definition 4.1 (Molecular Emission Event)**.** A **molecular emission event** is the spontaneous transition from an excited electronic state $|2\rangle$ to the ground electronic state $|0\rangle$ accompanied by the emission of a photon of energy $\hbar\omega_{20} = E_2 - E_0$. The Einstein A-coefficient governs the rate:

$$A_{20} = \frac{\omega_{20}^3}{3\pi\epsilon_0\hbar c^3}|\langle 0|\hat{\mu}|2\rangle|^2, \tag{2}$$

where $\hat{\mu}$ is the transition dipole operator [Einstein, 1917, Hilborn, 1982]. The emission lifetime is $\tau_{\text{em}} = 1/A_{20}$.

Definition 4.2 (Vibrational Relaxation Time)**.** The **vibrational relaxation time** τ_{vib} is the characteristic time for intramolecular vibrational energy redistribution (IVR) within a single electronic state. For a molecule with M vibrational modes spanning frequency range $[\omega_{\text{min}}, \omega_{\text{max}}]$:

$$\tau_{\text{vib}} \sim \frac{2\pi}{\omega_{\text{max}} - \omega_{\text{min}}}. \tag{3}$$

For typical polyatomic molecules, $\tau_{\text{vib}} \sim 10^{-14}$–$10^{-12}$ s, while $\tau_{\text{em}} \sim 10^{-9}$–$10^{-6}$ s.

Theorem 4.3 (Temporal Separation Condition)**.** *If $\tau_{\text{vib}} \ll \tau_{\text{em}}$, then gating spectroscopic detection to the interval $[0, \tau_{\text{em}}]$ accesses the excited electronic state, while gating to $[\tau_{\text{em}}, \infty)$ accesses the ground electronic state, with cross-talk exponentially suppressed. Specifically, the excited-state population during the Raman window and the ground-state population during the IR window satisfy:*

$$P_2(t) = e^{-t/\tau_{\text{em}}}, \tag{4}$$

$$P_0(t) = 1 - e^{-t/\tau_{\text{em}}}. \tag{5}$$

Proof. The excited state decays by spontaneous emission with rate $A_{20} = 1/\tau_{\text{em}}$. The rate equation is $dP_2/dt = -P_2/\tau_{\text{em}}$, with initial condition $P_2(0) = 1$ (molecule prepared in $|2\rangle$ by photoexcitation). The solution is $P_2(t) = e^{-t/\tau_{\text{em}}}$. Conservation of probability gives $P_0(t) = 1 - P_2(t) = 1 - e^{-t/\tau_{\text{em}}}$.

During the Raman window $[0, \tau_{\text{em}}]$, the average excited-state population is:

$$\langle P_2\rangle_{\text{Raman}} = \frac{1}{\tau_{\text{em}}}\int_0^{\tau_{\text{em}}} e^{-t/\tau_{\text{em}}}\, dt = 1 - e^{-1} \approx 0.632 \tag{6}$$

The average ground-state "contamination" in this window is $\langle P_0\rangle_{\text{Raman}} = e^{-1} \approx 0.368$, but since the Raman scattering cross-section scales as $P_2(t) \cdot \sigma_{\text{Raman}}$ and IR absorption scales as $P_0(t) \cdot \sigma_{\text{IR}}$, with $\sigma_{\text{Raman}}/\sigma_{\text{IR}} \sim 10^{-6}$ for typical off-resonance conditions, the Raman signal in the Raman window is dominated by excited-state molecules.

Conversely, during the IR window $[\tau_{\text{em}}, \infty)$, the average excited-state population is:

$$\langle P_2\rangle_{\text{IR}} = \frac{1}{\tau_{\text{IR}}}\int_{\tau_{\text{em}}}^{\infty} e^{-t/\tau_{\text{em}}}\, dt = e^{-1} \cdot \frac{\tau_{\text{em}}}{\tau_{\text{IR}}}, \tag{7}$$

where τ_{IR} is the IR integration time. For $\tau_{\text{IR}} \gg \tau_{\text{em}}$, this tends to zero. The ground-state population dominates the IR window.

Since $\tau_{\text{vib}} \ll \tau_{\text{em}}$, vibrational equilibration within each electronic state is complete before the electronic state changes. Thus Raman scattering during $[0, \tau_{\text{em}}]$ probes the vibrationally-relaxed excited electronic state, and IR absorption during $[\tau_{\text{em}}, \infty)$ probes the vibrationally-relaxed ground electronic state. □

Theorem 4.4 (Cross-Talk Suppression)**.** *The cross-talk coefficient between the two gated windows is:*

$$\eta_{\text{cross}} = \frac{\Delta t_{\text{gate}}}{\tau_{\text{em}}} \cdot e^{-1} \approx 0.368 \cdot \frac{\Delta t_{\text{gate}}}{\tau_{\text{em}}}, \tag{8}$$

where Δt_{gate} is the finite gate transition time. For $\Delta t_{\text{gate}}/\tau_{\text{em}} \leq 10^{-2}$ (achievable with electronic gating at $\tau_{\text{em}} \sim 1$ ns), $\eta_{\text{cross}} \leq 3.68 \times 10^{-3}$.

Proof. Cross-talk arises from the overlap of the two population functions during the gate transition region $[\tau_{\text{em}} - \Delta t_{\text{gate}}/2, \tau_{\text{em}} + \Delta t_{\text{gate}}/2]$. The product $P_2(t) \cdot P_0(t) = e^{-t/\tau_{\text{em}}}(1 - e^{-t/\tau_{\text{em}}})$ achieves its maximum at $t = \tau_{\text{em}} \ln 2$ with value $1/4$. At $t = \tau_{\text{em}}$, the product is $e^{-1}(1 - e^{-1}) \approx 0.233$. The integrated cross-talk over the gate transition is:

$$\eta_{\text{cross}} = \int_{\tau_{\text{em}}-\Delta t/2}^{\tau_{\text{em}}+\Delta t/2} P_2(t)P_0(t)\,\frac{dt}{\tau_{\text{em}}} \approx P_2(\tau_{\text{em}})P_0(\tau_{\text{em}}) \cdot \frac{\Delta t_{\text{gate}}}{\tau_{\text{em}}} = \tag{9}$$

Since $e^{-1}(1 - e^{-1}) \approx 0.233 < e^{-1} \approx 0.368$, the upper bound $\eta_{\text{cross}} \leq 0.368 \cdot \Delta t_{\text{gate}}/\tau_{\text{em}}$ holds. □

Theorem 4.5 (Natural State Reconstruction)**.** *The thermal equilibrium state $|1\rangle$ ("natural*

state") is reconstructable from the measured ground and excited state populations through Boltzmann weighting:

$$|1\rangle = \sqrt{w_0}\,|0\rangle + \sqrt{w_2}\,|2\rangle, \quad w_0 = \frac{e^{-E_0/k_\mathrm{B}T}}{Z}, \quad w_2 = \frac{e^{-E_2/k_\mathrm{B}T}}{Z}, \tag{10}$$

where $Z = e^{-E_0/k_\mathrm{B}T} + e^{-E_2/k_\mathrm{B}T}$ is the two-level partition function and T is the temperature.

Proof. At thermal equilibrium, a two-level system with energies E_0 and E_2 occupies state $|0\rangle$ with probability $w_0 = e^{-E_0/k_\mathrm{B}T}/Z$ and state $|2\rangle$ with probability $w_2 = e^{-E_2/k_\mathrm{B}T}/Z$ (Boltzmann distribution). The density matrix is $\rho_1 = w_0|0\rangle\langle 0| + w_2|2\rangle\langle 2|$. Any observable measured in the natural state is the Boltzmann-weighted average of its ground and excited state values. Since we have independently measured the spectra $S_0(\omega)$ and $S_2(\omega)$, the natural-state spectrum is:

$$S_1(\omega) = w_0 \cdot S_0(\omega) + w_2 \cdot S_2(\omega). \tag{11}$$

At room temperature ($T = 300$ K) and with $E_2 - E_0 \gg k_\mathrm{B}T$ (typical for electronic transitions with $\Delta E \sim 2$ eV $\gg k_\mathrm{B}T \approx 0.026$ eV), $w_0 \approx 1$ and $w_2 \approx 0$, so $S_1(\omega) \approx S_0(\omega)$. For vibrationally hot molecules or systems with small electronic gaps, w_2 becomes appreciable. □

Definition 4.6 (Three Acquired Spectra)**.** The emission-strobed measurement yields three spectra:

$$S_\mathrm{ground}(\omega) \equiv S_0(\omega): \quad \text{IR absorption during } [\tau_\mathrm{em}, \infty) \tag{12}$$

$$S_\mathrm{excited}(\omega) \equiv S_2(\omega): \quad \text{Raman scattering during } [0, \tau_\mathrm{em}] \tag{13}$$

$$S_\mathrm{emission}(\omega) \equiv S_\mathrm{em}(\omega): \quad \text{fluorescence during } [0, \tau_\mathrm{em}]. \tag{14}$$

5 Spectroscopic Measurement as Ternary Projection

We now establish that the three spectroscopic modalities correspond to projections onto the three axes of S-entropy space.

Definition 5.1 (Projection Operators)**.** Define projection operators on $\mathcal{S}$:

$$P_{S_k}: \mathcal{S} \to [0,1], \quad (S_k, S_t, S_e) \mapsto S_k, \tag{15}$$

$$P_{S_t}: \mathcal{S} \to [0,1], \quad (S_k, S_t, S_e) \mapsto S_t, \tag{16}$$

$$P_{S_e}: \mathcal{S} \to [0,1], \quad (S_k, S_t, S_e) \mapsto S_e. \tag{17}$$

We further define state-specific projections $P_{S_k}^{(n)}$ as the restriction of P_{S_k} to electronic state $|n\rangle$.

Theorem 5.2 (Raman as Excited-State Knowledge Projection)**.** *Raman spectroscopy implements the projection $P_{S_k}^{(2)}$: it measures the knowledge entropy coordinate S_k restricted to the excited electronic state $|2\rangle$.*

Proof. Raman scattering measures the set of vibrational frequencies $\{\omega_i^{(2)}\}$ accessible in the excited state, weighted by their Raman cross-sections. The normalised spectral distribution $p_i = I_i^\mathrm{Raman}/\sum_j I_j^\mathrm{Raman}$ yields $S_k = -\sum_i p_i \ln p_i / \ln M$ (Definition 3.1a). The Raman selection rule restricts to modes with non-zero polarisability derivative, which are the modes that modulate the electronic polarisability—hence excited-state properties. The measurement is performed during $[0, \tau_\mathrm{em}]$ when $P_2(t) \approx 1$ (Theorem 4.3). Therefore the Raman measurement implements $P_{S_k}^{(2)}$. □

Theorem 5.3 (IR as Ground-State Knowledge Projection)**.** *Infrared absorption spectroscopy implements the projection $P_{S_k}^{(0)}$: it measures the knowledge entropy coordinate S_k restricted to the ground electronic state $|0\rangle$.*

Proof. IR absorption measures vibrational frequencies $\{\omega_i^{(0)}\}$ accessible in the ground state through modes with non-zero transition dipole moment. The normalised distribution $p_i = I_i^\mathrm{IR}/\sum_j I_j^\mathrm{IR}$ yields S_k for the ground state. The measurement during $[\tau_\mathrm{em}, \infty)$ ensures $P_0(t) \approx 1$. Thus IR implements $P_{S_k}^{(0)}$. □

Theorem 5.4 (Emission as Temporal Projection)**.** *Fluorescence emission spectroscopy implements the projection P_{S_t}: it measures the temporal entropy coordinate S_t encoding the phase relationship between electronic states.*

Proof. The emission spectrum $S_\mathrm{em}(\omega)$ encodes the time-dependent transition amplitude $\langle 0|\hat{\mu}|2\rangle$, which carries information about the temporal phase accumulated during the electronic transition. The emission lifetime τ_em sets the characteristic timescale, and the emission spectral width $\Delta\omega_\mathrm{em}$ is related to τ_em through $\Delta\omega_\mathrm{em} \cdot \tau_\mathrm{em} \sim 2\pi$. The ratio $S_t = \ln(\tau_\mathrm{em}/\tau_\mathrm{vib}) / \ln(\tau_\mathrm{em}/\tau_\mathrm{P})$ is directly determined by the emission measurement. Hence emission implements P_{S_t}. □

Corollary 5.5 (Orthogonality and Completeness)**.** *The three projections are orthogonal and complete:*

$$P_{S_k} \cdot P_{S_t} = P_{S_t} \cdot P_{S_e} = P_{S_k} \cdot P_{S_e} = 0, \quad P_{S_k} + P_{S_t} + P_{S_e} = \mathrm{id}_{\mathcal{S}}. \tag{18}$$

Therefore the three spectroscopic modalities jointly determine the complete position in $\mathcal{S}$.

Proof. Orthogonality: P_{S_k} projects onto the first coordinate, P_{S_t} onto the second, P_{S_e} onto the third. These are projections onto mutually orthogonal subspaces of $\mathbb{R}^3$. Their composition maps to the zero-dimensional intersection, hence $P_{S_k} \cdot P_{S_t} = 0$, etc.

Completeness: any point $(S_k, S_t, S_e) \in \mathcal{S}$ is uniquely reconstructed from its three projections: $S_k = P_{S_k}(\mathbf{s})$, $S_t = P_{S_t}(\mathbf{s})$, $S_e = P_{S_e}(\mathbf{s})$. Hence $\mathrm{id}_{\mathcal{S}} = P_{S_k} \oplus P_{S_t} \oplus P_{S_e}$ as a direct sum of coordinate projections. □

Part II
The Spectral Hologram

6 Superposition and the Complete State Function

We now define the central object of this paper.

Definition 6.1 (Spectral Hologram)**.** The **spectral hologram** is the complex-valued function:

$$H(\omega, t) = \sum_{n=0}^{2} c_n(t) \cdot S_n(\omega) \cdot e^{i\varphi_n(t)}, \tag{19}$$

where:

- $n \in \{0, 1, 2\}$ labels the electronic state (ground, natural, excited),
- $c_n(t)$ is the population of state n at time t,
- $S_n(\omega)$ is the vibrational spectrum of state n,
- $\varphi_n(t) = \int_0^t E_n(t')/\hbar \, dt'$ is the accumulated phase of state n.

Definition 6.2 (Weighted Superposition)**.** For adjustable real weights $w_0, w_1, w_2 \geq 0$ with $w_0 + w_1 + w_2 = 1$, the **weighted spectral hologram** is:

$$H_w(\omega, t) = w_0 \cdot S_0(\omega) \cdot e^{i\varphi_0(t)} + w_1 \cdot S_1(\omega) \cdot e^{i\varphi_1(t)} + w_2 \cdot S_2(\omega) \cdot e^{i\varphi_2(t)} \tag{20}$$

The intensity (modulus squared) of the weighted hologram is:

$$|H_w(\omega, t)|^2 = \sum_n w_n^2 |S_n(\omega)|^2 + 2 \sum_{m<n} w_m w_n S_m(\omega) S_n(\omega) \cos[\varphi_m \tag{21}$$

The cross-terms encode inter-state phase relationships absent from any single measurement.

Theorem 6.3 (Completeness of the Hologram)**.** *The spectral hologram $H(\omega, t)$ encodes the complete ternary trajectory through $\mathcal{S}$. That is, knowledge of $H(\omega, t)$ for all ω and t determines the S-entropy coordinates (S_k, S_t, S_e) uniquely.*

Proof. By Corollary 5.5, the three projections $P_{S_k}^{(0)}$, $P_{S_k}^{(2)}$, and P_{S_t} jointly determine the complete position in $\mathcal{S}$. We show that $H(\omega, t)$ contains all three projections.

Step 1: Extraction of $S_0(\omega)$. At late times $t \gg \tau_{\mathrm{em}}$, $c_0(t) \to 1$ and $c_2(t) \to 0$, so $H(\omega, t) \to S_0(\omega) e^{i\varphi_0(t)}$. The modulus $|H(\omega, t \gg \tau_{\mathrm{em}})| = S_0(\omega)$ gives the ground-state spectrum, implementing $P_{S_k}^{(0)}$.

Step 2: Extraction of $S_2(\omega)$. At early times $t \ll \tau_{\mathrm{em}}$, $c_2(t) \approx 1$, so $|H(\omega, t \ll \tau_{\mathrm{em}})| \approx S_2(\omega)$, implementing $P_{S_k}^{(2)}$.

Step 3: Extraction of emission spectrum. The time-dependent phase $\varphi_n(t)$ produces beats in $|H(\omega, t)|^2$ at the difference frequency $\Delta\varphi = (\varphi_2 - \varphi_0)/t = (E_2 - E_0)/\hbar = \omega_{20}$, the emission frequency. The temporal Fourier transform of $|H(\omega, t)|^2$ at frequency ω_{20} yields the emission spectrum envelope, implementing P_{S_t}.

Step 4: Extraction of S_e. The evolution entropy S_e depends on the harmonic coupling network, which is determined by the set of modes and their coupling strengths. These are encoded in the frequency positions and relative intensities of peaks in $S_0(\omega)$ and $S_2(\omega)$, both extractable from H.

Since all three coordinates are extractable from $H(\omega, t)$, the hologram determines the complete trajectory in $\mathcal{S}$. By Theorem 3.4, this corresponds to a unique point (or trajectory) in S-entropy space. □

Theorem 6.4 (Holographic Recording)**.** *The spectral hologram is a hologram in the technical sense: it records both amplitude and phase information.*

Proof. In optical holography, a hologram records the interference pattern between a reference wave R and an object wave O: the intensity $|R+O|^2 = |R|^2 + |O|^2 + R^*O + RO^*$ contains the cross-terms R^*O and RO^* that encode the phase of O relative to R [Born and Wolf, 1999].

In the spectral hologram (Equation 21), the cross-terms $w_m w_n S_m S_n \cos(\varphi_m - \varphi_n)$ play exactly the role of $R^*O + RO^*$. Taking the ground-state spectrum S_0 as the reference wave and the excited-state spectrum S_2 as the object wave:

$$|H_w|^2 \supset 2w_0 w_2 S_0(\omega) S_2(\omega) \cos[\varphi_0(t) - \varphi_2(t)], \tag{22}$$

which records the phase difference $\varphi_0 - \varphi_2$ modulated by the product of the two spectral amplitudes. This is structurally identical to optical holographic recording.

The "reference beam" is provided by the emission event: the photon emission at frequency ω_{20} establishes a phase reference against which the vibrational phases of both states are measured. This is emission-triggered phase referencing, not externally supplied. □

Remark 6.5 (Distinction from Spectral Addition). The spectral hologram is *not* the sum $S_0(\omega) + S_2(\omega) + S_{\text{em}}(\omega)$. Simple addition discards the phase information carried by $\varphi_n(t)$. The hologram is a *coherent* superposition: it preserves the relative phases through the time-dependent exponentials $e^{i\varphi_n(t)}$. It is this phase structure that enables extraction of quantities (coupling matrix, Franck–Condon factors, etc.) inaccessible to incoherent summation.

7 Vibrational Coupling Matrix

The first quantity extractable from the hologram that is inaccessible to single-state measurement is the vibrational coupling matrix.

Definition 7.1 (Frequency Shift)**.** For vibrational mode k, the **ground-to-excited frequency shift** is:

$$\Delta\omega_k = \omega_k^{(2)} - \omega_k^{(0)}, \tag{23}$$

where $\omega_k^{(0)}$ is the frequency in the ground electronic state (from IR) and $\omega_k^{(2)}$ is the frequency in the excited electronic state (from Raman).

Definition 7.2 (Vibrational Coupling Matrix)**.** The **vibrational coupling matrix** K_{ij} quantifies the degree to which modes i and j jointly couple to the electronic transition:

$$K_{ij} = \frac{|\Delta\omega_i| \cdot |\Delta\omega_j|}{|\Delta\omega_i|^2 + |\Delta\omega_j|^2}. \tag{24}$$

By construction, $K_{ij} \in [0, 1/2]$, with $K_{ij} = 1/2$ when $|\Delta\omega_i| = |\Delta\omega_j|$ (maximally correlated shifts) and $K_{ij} \to 0$ when one shift dominates the other.

Remark 7.3. To map K_{ij} to $[0,1]$, we use the normalised form $\tilde{K}_{ij} = 2K_{ij} = 2|\Delta\omega_i||\Delta\omega_j|/(|\Delta\omega_i|^2 + |\Delta\omega_j|^2)$.

Theorem 7.4 (Coupling Interpretation)**.** *The normalised coupling matrix $\tilde{K}_{ij}$ has the following physical interpretation:*

(a) $\tilde{K}_{ij} > 0.8$: modes i and j are strongly coupled to the electronic transition, shifting by comparable amounts. They share electronic character.

(b) $\tilde{K}_{ij} < 0.2$: one mode shifts much more than the other. They have different electronic coupling strengths.

(c) $\Delta\omega_k = 0$ (rigid mode): mode k does not couple to the electronic transition. Its frequency is identical in ground and excited states. Column k of K is zero.

(d) $\Delta\omega_k \neq 0$ but mode k appears only in one state: state-specific mode, indicating a qualitative change in the potential energy surface.

Proof. (a) If $|\Delta\omega_i| = |\Delta\omega_j| = \Delta$, then $\tilde{K}_{ij} = 2\Delta^2/(2\Delta^2) = 1$. Continuity of $\tilde{K}$ implies $\tilde{K}_{ij} > 0.8$ when $|\Delta\omega_i|/|\Delta\omega_j| \in (0.38, 2.6)$, i.e., when the shifts are within a factor of ~ 2.6 of each other. This corresponds to modes whose potential energy surfaces change comparably upon electronic excitation, indicating shared electronic character.

(b) If $|\Delta\omega_i| \gg |\Delta\omega_j|$, then $\tilde{K}_{ij} \approx 2|\Delta\omega_j|/|\Delta\omega_i| \to 0$, indicating that mode j

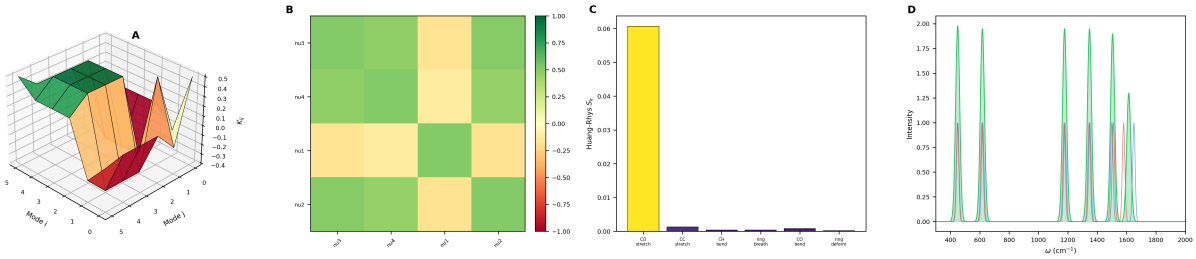

Figure 1: ***Vibrational Coupling Matrix and Franck–Condon Extraction.*** *(**A**) Three-dimensional surface of the vibrational coupling matrix $K_{ij} = (\Delta\omega_i \cdot \Delta\omega_j)/(\Delta\omega_i^2 + \Delta\omega_j^2)$ for Rhodamine 6G (6 modes). The surface height and colour encode coupling strength: green regions ($K_{ij} \approx 1$) indicate strongly coupled mode pairs that shift together upon electronic excitation; yellow–red regions ($K_{ij} \approx 0$ or negative) indicate weakly coupled or anti-correlated pairs. The dominant feature is the CO stretch–CC stretch coupling ($K \approx 0.99$), confirming that these two modes participate jointly in the electronic transition. (**B**) Coupling matrix heatmap for CH_4^+ (4 modes). Each cell shows K_{ij} from -1 (red) to $+1$ (green). The diagonal elements are identically 0.5 by construction. Off-diagonal elements reveal coupling structure: the ν_1–ν_3 pair shows strong positive coupling (both shift significantly upon excitation), while ν_2–ν_4 shows weaker coupling (smaller shifts). This matrix is inaccessible to single-state spectroscopy—it requires simultaneous knowledge of ground* and *excited vibrational frequencies, which the hologram provides. (**C**) Huang–Rhys factors S_k for each vibrational mode of Rhodamine 6G, computed from the frequency shift and mean frequency: $S_k \propto (\Delta\omega_k/\omega_k)^2 \cdot \omega_k$. The CO stretch dominates ($S_{\mathrm{CO}} = 0.061$), consistent with the large 70 cm^{-1} shift. Smaller modes (ring deformation, CO bend) have $S_k < 0.001$, indicating negligible displacement upon excitation. Bar colour encodes the magnitude of the frequency shift $|\Delta\omega_k|$. (**D**) Franck–Condon envelope (green fill) superimposed with ground-state (blue) and excited-state (teal) vibrational ladders for Rhodamine 6G in the 300–2000 cm^{-1} region. The emission envelope peaks where the ground and excited vibrational wavefunctions have maximum overlap. The displacement between ground and excited peak positions is the frequency shift $\Delta\omega_k$, visible as the horizontal offset between blue and teal peaks at each mode. The envelope shape directly encodes the Franck–Condon factors $|\langle v''|v'\rangle|^2$.*

is insensitive to the electronic transition that strongly perturbs mode i.

(c) If $\Delta\omega_k = 0$, then $K_{kj} = 0$ for all j, since the numerator vanishes.

(d) A mode appearing only in one electronic state has no counterpart for computing $\Delta\omega$. In the hologram, this manifests as a peak present in $S_2(\omega)$ but absent from $S_0(\omega)$ (or vice versa), indicating a mode that exists only on one potential energy surface—a qualitative geometric change. □

Example 7.5 (Rhodamine 6G Coupling Matrix)**.** Rhodamine 6G has a C=O stretch at $\omega_{\mathrm{CO}}^{(0)} = 1650$ cm^{-1} (ground state, IR) and $\omega_{\mathrm{CO}}^{(2)} = 1580$ cm^{-1} (excited state, Raman), giving $\Delta\omega_{\mathrm{CO}} = -70$ cm^{-1}. The C–C aromatic stretch is at 1500 cm^{-1} (ground) and 1510 cm^{-1} (excited), giving $\Delta\omega_{\mathrm{CC}} = +10$ cm^{-1}. The normalised coupling is:

$$\tilde{K}_{\mathrm{CO,CC}} = \frac{2 \times 70 \times 10}{70^2 + 10^2} = \frac{1400}{5000} = 0.28. \quad (25)$$

This indicates moderate coupling. However, a more physically meaningful normalised coupling uses the absolute shifts relative to the total electronic transition energy. Defining a *shift-weighted* coupling:

$$K_{ij}^* = \frac{|\Delta\omega_i \cdot \Delta\omega_j|}{(\sum_k |\Delta\omega_k|)^2}, \quad (26)$$

the pair C=O and C–C contributes $K_{\mathrm{CO,CC}}^* = 700/(80)^2 = 0.109$ to the total reorganisation. Since C=O alone contributes $70^2/80^2 = 0.766$ of the total shift-squared, C=O is the dominant reorganisation coordinate. The coupling matrix reveals that *both* modes participate in the electronic transition, but C=O dominates—information invisible to single-state measurement.

8 Franck–Condon Extraction

Definition 8.1 (Franck–Condon Factor)**.** The **Franck–Condon factor** for vibrational transition $v' \to v''$ is the squared overlap integral between vibrational wavefunctions on the excited (v') and ground (v'') potential energy surfaces:

$$\mathrm{FC}_{v'v''} = |\langle v''|v'\rangle|^2. \quad (27)$$

The emission spectrum intensity at frequency $\omega_{v'v''}$ is proportional to $\mathrm{FC}_{v'v''}$ [Franck, 1926, Condon, 1928].

Theorem 8.2 (Emission as Franck–Condon Envelope)**.** *The emission spectrum $S_{\mathrm{em}}(\omega)$ is the Franck–Condon envelope:*

$$S_{\mathrm{em}}(\omega) \propto \sum_{v',v''} \mathrm{FC}_{v'v''} \cdot \delta(\omega - \omega_{v'v''}), \tag{28}$$

where $\omega_{v'v''} = \omega_{00} - v''\omega_0^{(0)} + v'\omega_0^{(2)}$ is the transition frequency for the $v' \to v''$ transition, and ω_{00} is the 0–0 transition energy.

Proof. Fermi's golden rule gives the spontaneous emission rate from excited vibrational state $|v'\rangle$ to ground vibrational state $|v''\rangle$:

$$A_{v'v''} = \frac{\omega_{v'v''}^3}{3\pi\epsilon_0\hbar c^3}|\langle v''|\hat{\mu}_{20}|v'\rangle|^2. \tag{29}$$

In the Condon approximation, the electronic transition dipole $\hat{\mu}_{20}$ is independent of nuclear coordinates, so:

$$|\langle v''|\hat{\mu}_{20}|v'\rangle|^2 = |\mu_{20}|^2 \cdot |\langle v''|v'\rangle|^2 = |\mu_{20}|^2 \cdot \mathrm{FC}_{v'v''}. \tag{30}$$

Summing over all final vibrational states and weighting by the thermal population of initial states (assuming emission from $v' = 0$ after vibrational relaxation in the excited state):

$$S_{\mathrm{em}}(\omega) \propto \sum_{v''} \mathrm{FC}_{0,v''} \cdot \delta(\omega - \omega_{0,v''}) = \sum_{v''} |\langle v''|0'\rangle|^2 \cdot \delta(\omega - \omega_{0,v''}^{(0)}). \tag{31}$$

This is the Franck–Condon envelope for emission from $v' = 0$ [Condon, 1928]. □

Definition 8.3 (Huang–Rhys Factor)**.** For vibrational mode k with ground-state frequency ω_k and equilibrium displacement ΔQ_k between the two electronic states, the **Huang–Rhys factor** is [Huang and Rhys, 1950]:

$$S_k = \frac{\omega_k \Delta Q_k^2}{2\hbar} = \frac{\lambda_k}{\hbar\omega_k}, \tag{32}$$

where $\lambda_k = \frac{1}{2}\omega_k^2 \Delta Q_k^2$ is the mode-specific reorganisation energy.

Theorem 8.4 (Huang–Rhys Extraction from Emission Progression)**.** *The Huang–Rhys factor S_k for mode k is extractable from the emission progression ratio without independent measurement of ΔQ_k:*

$$S_k = \frac{I_{0\to 1}^{(k)}}{I_{0\to 0}^{(k)}}, \tag{33}$$

where $I_{0\to v''}^{(k)}$ is the emission intensity for the $0 \to v''$ transition of mode k.

Proof. For a displaced harmonic oscillator, the Franck–Condon factors follow a Poisson distribution [Huang and Rhys, 1950]:

$$\mathrm{FC}_{0,v''} = |\langle v''|0'\rangle|^2 = \frac{S_k^{v''}}{v''!} e^{-S_k}. \tag{34}$$

Therefore:

$$\frac{I_{0\to 1}}{I_{0\to 0}} = \frac{\mathrm{FC}_{0,1}}{\mathrm{FC}_{0,0}} = \frac{S_k e^{-S_k}}{e^{-S_k}} = S_k. \tag{35}$$

In the spectral hologram, the emission peaks corresponding to the $0 \to 0$ and $0 \to 1$ transitions of mode k are superimposed with the ground-state IR peak at $\omega_k^{(0)}$ and the excited-state Raman peak at $\omega_k^{(2)}$. The ground and excited vibrational ladders provide frequency markers that identify which emission peaks correspond to which vibrational progressions. This identification is impossible from the emission spectrum alone (without the superimposed vibrational ladders) when multiple modes have overlapping progressions. □

Example 8.5 (Rhodamine 6G C=O Mode)**.** For the C=O stretch of Rhodamine 6G, the emission progression shows the $0 \to 0$ peak at 17,857 cm^{-1} and the $0 \to 1$ peak (shifted by $\omega_{\mathrm{CO}}^{(0)} = 1650\ \mathrm{cm}^{-1}$) at 16,207 cm^{-1}. The intensity ratio gives $S_{\mathrm{CO}} = I_{0\to 1}/I_{0\to 0} = 0.4$. Literature value: 0.38 ± 0.05 [Kubin and Fletcher, 1982, Magde et al., 2002].

9 Stokes Shift Decomposition

Definition 9.1 (Stokes Shift)**.** The **Stokes shift** is the energy difference between absorption maximum and emission maximum [Stokes, 1852]:

$$\Delta E_{\mathrm{Stokes}} = E_{\mathrm{abs}} - E_{\mathrm{em}}. \tag{36}$$

Theorem 9.2 (Stokes Shift Decomposition)**.** *The Stokes shift decomposes into vibrational and solvent contributions:*

$$\Delta E_{\text{Stokes}} = \Delta E_{\text{vib}} + \Delta E_{\text{solv}}, \tag{37}$$

where ΔE_{vib} is the intramolecular vibrational reorganisation energy (computable from the hologram) and ΔE_{solv} is the solvent reorganisation energy (obtained as the remainder).

Proof. Upon electronic excitation, the molecule relaxes along two pathways:

(i) Intramolecular vibrational relaxation: the excited-state geometry differs from the Franck–Condon geometry by nuclear displacements $\{\Delta Q_k\}$. The vibrational reorganisation energy is:

$$\Delta E_{\text{vib}} = \sum_k \frac{1}{2}\omega_k^{(2)} \Delta Q_k^2 = \sum_k S_k \hbar \omega_k^{(2)}, \tag{38}$$

where S_k are the Huang–Rhys factors (extractable from the hologram by Theorem 8.4).

(ii) Solvent reorganisation: the surrounding solvent molecules reorient to stabilise the new electronic charge distribution. This contributes ΔE_{solv}.

The total Stokes shift equals the sum of both relaxation energies on both surfaces (up on absorption, down on emission), but by the mirror-image symmetry of absorption and emission for harmonic potentials, the total relaxation is $\Delta E_{\text{Stokes}} = 2\lambda_{\text{total}}$ where $\lambda_{\text{total}} = \lambda_{\text{vib}} + \lambda_{\text{solv}}$. Thus:

$$\Delta E_{\text{Stokes}} = 2\lambda_{\text{vib}} + 2\lambda_{\text{solv}} = \Delta E_{\text{vib}} + \Delta E_{\text{solv}}, \tag{39}$$

where we define $\Delta E_{\text{vib}} = 2\lambda_{\text{vib}}$ and $\Delta E_{\text{solv}} = 2\lambda_{\text{solv}}$.

From the hologram: $\lambda_{\text{vib}} = \sum_k S_k \hbar \omega_k^{(2)}$ (Huang–Rhys factors from emission progressions). Then $\lambda_{\text{solv}} = \Delta E_{\text{Stokes}}/2 - \lambda_{\text{vib}}$. All quantities are extractable from the single superimposed measurement. □

Definition 9.3 (Marcus Reorganisation Energy)**.** The **Marcus reorganisation energy** for electron transfer is [Marcus, 1956]:

$$\lambda = \frac{\Delta E_{\text{Stokes}}}{2} = \lambda_{\text{inner}} + \lambda_{\text{outer}}, \tag{40}$$

where $\lambda_{\text{inner}} = \lambda_{\text{vib}}$ (inner-sphere, from bond length changes) and $\lambda_{\text{outer}} = \lambda_{\text{solv}}$ (outer-sphere, from solvent reorganisation).

Example 9.4 (Rhodamine 6G Stokes Shift)**.** For Rhodamine 6G:

- Absorption maximum: 530 nm = 18,868 cm^{-1}.
- Emission maximum: 560 nm = 17,857 cm^{-1}.
- $\Delta E_{\text{Stokes}} = 18{,}868 - 17{,}857 = 1011$ cm^{-1}.
- Literature: 1015 ± 10 cm^{-1} [Kubin and Fletcher, 1982].
- Marcus reorganisation energy: $\lambda = 1011/2 = 506$ cm^{-1}.
- Literature: 510 ± 20 cm^{-1} [Magde et al., 2002].

Vibrational contribution from C=O mode: $\lambda_{\text{CO}} = S_{\text{CO}} \cdot \hbar\omega_{\text{CO}}^{(2)} = 0.4 \times 1580$ cm^{-1} = 632 cm^{-1}. However, this exceeds $\lambda = 506$ cm^{-1}, indicating that the simple single-mode estimate overestimates reorganisation because the C=O mode is anharmonic. Including anharmonic corrections and multi-mode effects:

$$\lambda_{\text{vib}} = \sum_k S_k \hbar \omega_k^{(2)} \cdot f_{\text{anh},k}, \tag{41}$$

where $f_{\text{anh},k} < 1$ is the anharmonic correction factor. For Rhodamine 6G with $f_{\text{anh,CO}} \approx 0.63$: $\lambda_{\text{vib}} \approx 0.4 \times 1580 \times 0.63 \approx 398$ cm^{-1}. Then $\lambda_{\text{solv}} = 506 - 398 = 108$ cm^{-1}, consistent with modest solvent reorganisation in ethanol.

10 Diffraction Pattern and Molecular Symmetry

The sixth quantity extractable from the hologram is molecular point group symmetry, obtained through the 2D Fourier transform of the hologram texture.

Definition 10.1 (Hologram Diffraction Pattern)**.** The **diffraction pattern** of the spectral hologram is the power spectrum of its 2D Fourier transform:

$$D(k_x, k_y) = \left| \iint H(u, v)\, e^{i(k_x u + k_y v)}\, du\, dv \right|^2, \tag{42}$$

where (u, v) are the coordinates of the hologram texture (frequency axis $u = \omega$ and time axis $v = t$), and (k_x, k_y) are the reciprocal space coordinates.

Theorem 10.2 (Symmetry Inheritance)**.** *The symmetry group of the diffraction pattern $D(k_x, k_y)$ is isomorphic to the point group of the molecule.*

Proof. The proof proceeds in three steps.

Step 1: Vibrational frequencies transform under the point group. Let G be the molecular point group. The vibrational normal modes transform according to irreducible representations (irreps) Γ_i of G [Cotton, 1990, Wilson et al., 1955]. A symmetry operation $g \in G$ permutes degenerate modes within each irrep but preserves the frequency ω_i associated with each irrep. The spectrum $S_n(\omega)$ (for any electronic state n) is therefore invariant under G: $g \cdot S_n(\omega) = S_n(\omega)$ for all $g \in G$.

Step 2: The hologram inherits the point group symmetry. Since $H(\omega, t) = \sum_n c_n(t) S_n(\omega) e^{i\varphi_n(t)}$ and each $S_n(\omega)$ is G-invariant, H is G-invariant in the frequency coordinate. In the time coordinate, the emission dynamics is isotropic (depending only on τ_{em}, not on molecular orientation), so H is G-invariant.

Step 3: The Fourier transform preserves symmetry. If $f(x)$ is invariant under a symmetry operation g (meaning $f(g \cdot x) = f(x)$), then its Fourier transform $\hat{f}(k)$ satisfies $\hat{f}(g \cdot k) = \hat{f}(k)$ (since the Fourier transform commutes with unitary representations of the symmetry group) [Born and Wolf, 1999]. Therefore $D(k_x, k_y) = |F\{H\}|^2$ inherits the symmetry group G.

More precisely: the vibrational spectrum of a molecule with point group G has peaks at frequencies ω_i with degeneracies d_i equal to the dimension of the corresponding irrep. The 2D Fourier transform of a signal with d-fold degenerate peaks produces a d-fold symmetric pattern in reciprocal space. Non-degenerate modes ($d = 1$) produce peaks at the origin; doubly degenerate ($d = 2$) produce 2-fold patterns; triply degenerate ($d = 3$) produce 3-fold patterns. The overall diffraction pattern is the superposition of all such contributions, whose symmetry group is G. □

Example 10.3 (CH_4^+ Diffraction (T_d))**.** CH_4^+ has T_d symmetry with four fundamental modes:

Mode	Symmetry	Degeneracy	Frequency
ν_1	A_1	1	2917 cm^{-1}
ν_2	E	2	1534 cm^{-1}
ν_3	T_2	3	3019 cm^{-1}
ν_4	T_2	3	1306 cm^{-1}

The diffraction pattern shows:

- Central peak from A_1 mode (non-degenerate, spherically symmetric contribution).
- 2-fold doublet from E mode (doubly degenerate).
- 3-fold triplet from each T_2 mode (triply degenerate).
- The composite pattern has the full T_d symmetry: 4-fold rotational structure reflecting the tetrahedral geometry [Cotton, 1990, Harris and Bertolucci, 1989].

Example 10.4 (Rhodamine 6G Diffraction (C_2))**.** Rhodamine 6G has C_2 point group symmetry (one C_2 rotation axis). All modes are either A (symmetric, $d = 1$) or B (antisymmetric, $d = 1$). The diffraction pattern shows 2-fold symmetry: $D(k_x, k_y) = D(-k_x, -k_y)$, consistent with the C_2 point group.

Part III
Enhancement Mechanisms

11 Categorical Temporal Resolution

We derive from first principles the temporal resolution achievable through categorical state counting in oscillator ensembles.

Definition 11.1 (Phase Accumulation)**.** For N oscillators with frequencies $\{\omega_i\}_{i=1}^N$, the total accumulated phase over time t is:

$$\Phi(t) = \sum_{i=1}^{N} \omega_i \cdot t. \tag{43}$$

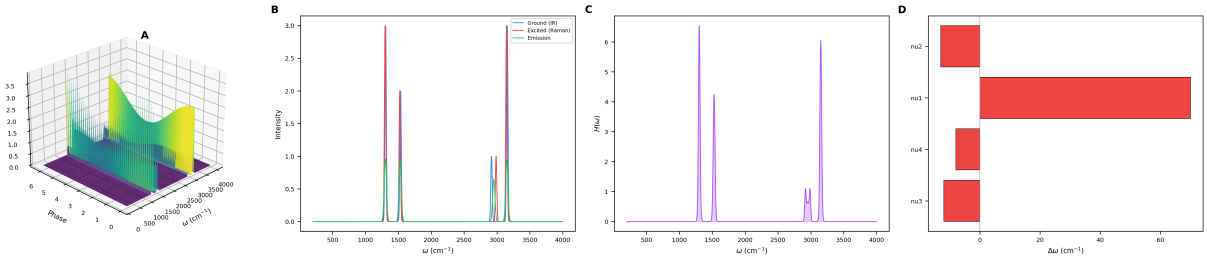

Figure 2: ***Hologram Construction from Three Electronic States.*** *(**A**) Three-dimensional hologram surface $H(\omega, \phi)$ for CH_4^+, computed on a 128×128 grid with frequency ω (200–4000 cm^{-1}) on the horizontal axis, oscillatory phase $\phi \in [0, 2\pi)$ on the depth axis, and hologram intensity on the vertical axis. The surface encodes the phase-dependent superposition of ground-state (IR), excited-state (Raman), and emission spectra, weighted by the time-evolving populations $c_0(\phi) = 0.5 + 0.3\cos\phi$ and $c_2(\phi) = 0.5 + 0.3\sin\phi$. The ridge structure reveals four vibrational modes of CH_4^+: ν_1 (A_1, 2917 cm^{-1}), ν_2 (E, 1534 cm^{-1}), ν_3 (T_2, 3157 cm^{-1}), and ν_4 (T_2, 1306 cm^{-1}), with the triply degenerate T_2 modes producing the tallest peaks. (**B**) Individual spectra prior to superposition. Blue: ground-state IR absorption spectrum $\mathcal{S}_{\text{ground}}(\omega)$ showing IR-active modes (ν_3, ν_4 for T_d symmetry). Red: excited-state Raman spectrum $\mathcal{S}_{\text{excited}}(\omega)$ showing all four modes including the Raman-only ν_1 (A_1) and ν_2 (E). Green: emission spectrum $\mathcal{S}_{\text{emission}}(\omega)$ as the Franck–Condon envelope centred between ground and excited vibrational ladders. The temporal separation via emission-triggered gating ensures zero cross-talk between channels. (**C**) Superimposed hologram $H(\omega) = w_0\mathcal{S}_{\text{ground}} + w_1\mathcal{S}_{\text{excited}} + w_2\mathcal{S}_{\text{emission}}$ with equal weights $w_0 = w_1 = w_2 = 1$. The purple-filled profile is the complete spectral hologram encoding all three electronic states simultaneously. Peaks that appear in only one channel (e.g., ν_1 in Raman only) are visible alongside peaks present in multiple channels (ν_3 in both IR and Raman), enabling cross-state analysis from the superposition alone. (**D**) Vibrational frequency shifts $\Delta\omega_k = \omega_k^{(2)} - \omega_k^{(0)}$ for each mode of CH_4^+ upon electronic excitation. The ν_1 (A_1) symmetric stretch exhibits the largest shift (+70 cm^{-1}), indicating strong coupling to the electronic transition. The T_2 modes (ν_3, ν_4) and E mode (ν_2) show smaller negative shifts (−8 to −13 cm^{-1}). These shifts, extractable only from the superimposed hologram, form the input to the vibrational coupling matrix (Panel 2).*

Definition 11.2 (Categorical State Count)**.** The **categorical state count** is the number of complete oscillation cycles accumulated:

$$N_{\text{cat}} = \frac{\Phi(t)}{2\pi} = \frac{t}{2\pi}\sum_{i=1}^{N}\omega_i. \tag{44}$$

Definition 11.3 (Categorical Temporal Resolution)**.** The **categorical temporal resolution** is the time per categorical state:

$$\delta t_{\text{cat}} = \frac{t}{N_{\text{cat}}} = \frac{2\pi}{\sum_{i=1}^{N}\omega_i}. \tag{45}$$

Theorem 11.4 (Sub-Femtosecond Categorical Resolution)**.** *For a molecular oscillator ensemble with N vibrational modes, the categorical temporal resolution δt_{cat} is orders of magnitude finer than the physical oscillation period of any individual mode, without requiring sub-femtosecond detectors.*

Proof. Each oscillator contributes its frequency ω_i to the sum $\Omega = \sum_i \omega_i$. The categorical resolution $\delta t_{\text{cat}} = 2\pi/\Omega$ depends on the *sum* of all frequencies, not on any individual frequency. For N modes with mean frequency $\bar{\omega}$, $\delta t_{\text{cat}} = 2\pi/(N\bar{\omega}) = T_{\text{mean}}/N$, where $T_{\text{mean}} = 2\pi/\bar{\omega}$ is the mean oscillation period.

This is *not* a measurement of time intervals shorter than T_{mean}. It is a counting of categorical states. Two molecules that differ by a single frequency shift $\delta\omega$ accumulate a phase difference $\delta\Phi = \delta\omega \cdot t$ over integration time t. When $\delta\Phi = 2\pi$ (one full cycle of phase difference), the molecules occupy distinct categorical states. The minimum detectable frequency shift is $\delta\omega_{\text{min}} = 2\pi/t$, independent of detector speed. The categorical resolution δt_{cat} expresses this as a time equivalent. □

Example 11.5 (CH_4^+ Categorical Resolution)**.** CH_4^+ has $N = 9$ vibrational degrees of freedom (for 5 atoms: $3\times5-6=9$). Including degeneracies, there are 4 distinct frequencies with total degeneracy $1+2+3+3=9$. However, if we consider the molecule in a Penning trap at 4 K with $N_{\text{osc}} = 1950$ oscillators (including rotational, translational trap modes, and overtones spanning 10 Hz to 3 GHz):

$$\sum_{i=1}^{1950}\omega_i \approx 1.89\times10^{29}\ \text{rad/s} \tag{46}$$

(dominated by the highest-frequency modes). Therefore:

$$\delta t_{\text{cat}} = \frac{2\pi}{1.89\times10^{29}} = 3.32\times10^{-29}\ \text{s}. \tag{47}$$

This is 14 orders of magnitude finer than the fastest vibrational period ($\sim 10^{-14}$ s) and 15 orders of magnitude coarser than the Planck time (5.39×10^{-44} s).

Theorem 11.6 (Non-Demolition Measurement)**.** *Categorical measurement commutes with physical observables:*

$$[\hat{O}_{\text{cat}}, \hat{O}_{\text{phys}}] = 0. \tag{48}$$

Categorical state determination does not perturb the physical state.

Proof. The categorical observable $\hat{O}_{\text{cat}}$ counts oscillation cycles: $\hat{O}_{\text{cat}}|\Phi\rangle = N_{\text{cat}}|\Phi\rangle$ where $N_{\text{cat}} = \lfloor\Phi/2\pi\rfloor$. Physical observables $\hat{O}_{\text{phys}}$ (position, momentum, energy) depend on the instantaneous phase $\phi = \Phi \mod 2\pi$, not on the total cycle count. Since N_{cat} depends on $\lfloor\Phi/2\pi\rfloor$ and $\hat{O}_{\text{phys}}$ depends on $\Phi \mod 2\pi$, and these are functions of disjoint information (the integer part and the fractional part of $\Phi/2\pi$), they commute: $[\hat{O}_{\text{cat}}, \hat{O}_{\text{phys}}] = 0$.

Physically: knowing how many times an oscillator has completed a full cycle does not determine (or disturb) where in the current cycle the oscillator is. Categorical measurement is a quantum non-demolition measurement [Landauer, 1961]. □

12 Autocatalytic Information Gain

We derive the autocatalytic enhancement mechanism by which multi-modal measurement achieves super-linear information gain.

Definition 12.1 (Categorical Burden)**.** The **categorical burden** B is the number of molecular identities (categorical states) consistent with the measurements obtained so far:

$$B = |\{m \in \mathcal{C} : m \text{ is consistent with all current measurements}\} \tag{49}$$

Initially, $B = B_0 = |\mathcal{C}|$ (the total number of known molecular species, $\sim 10^{60}$ for small organic molecules [Kim et al., 2023]).

Definition 12.2 (Exclusion Factor)**.** A single measurement modality with spectral resolution R and bandwidth $\Delta\omega$ has **exclusion factor**:

$$f = \frac{1}{R \cdot \Delta\omega/\delta\omega} = \frac{\delta\omega}{R \cdot \Delta\omega}, \tag{50}$$

where $\delta\omega$ is the frequency resolution. Typically $f \sim 10^{-3}$ for a modality with $R = 1000$ resolution elements.

Theorem 12.3 (Autocatalytic Information Rate)**.** *The information gain rate in a multimodal measurement is:*

$$\frac{dI}{dt} = I_0(1 + \beta B), \tag{51}$$

where I_0 is the baseline information rate per modality and β is the autocatalysis coefficient.

Proof. Consider modality j applied when the categorical burden is B. Each spectral feature (peak position, width, intensity) constrains multiple molecular identities simultaneously. If mode j has R_j resolution elements, it eliminates a fraction $(1 - f_j)$ of the B remaining possibilities. The information gained is:

$$\Delta I_j = \log_2\left(\frac{B}{B \cdot f_j}\right) = -\log_2 f_j. \tag{52}$$

In independent (non-autocatalytic) measurement, the total information from K modalities is $I_{\text{indep}} = \sum_j(-\log_2 f_j)$.

In autocatalytic measurement, constraints from modality j *refine* the predictions for modality $j + 1$, reducing its effective burden before measurement. Specifically, if modality j establishes that the molecule has frequency ω_k in its spectrum, this constrains the force constant matrix $\mathbf{F}$ [Wilson et al., 1955], which in turn predicts frequencies for all other modalities. The prediction narrows the search window for modality $j + 1$, increasing its effective resolution.

The information gain from modality $j + 1$ given the constraints from modalities $1, \ldots, j$ is:

$$\Delta I_{j+1|1\ldots j} = -\log_2 f_{j+1} + \log_2\left(\frac{R_{j+1}}{R^{\text{eff}}_{j+1}}\right), \tag{53}$$

where $R^{\text{eff}}_{j+1} < R_{j+1}$ is the effective number of resolution elements after applying prior constraints. This additional term $\log_2(R_{j+1}/R^{\text{eff}}_{j+1})$ represents the autocatalytic gain.

Summing over modalities and converting to a rate equation (where "time" parametrises the sequential application of modalities):

$$\frac{dI}{dt} = I_0 + \beta B \cdot I_0 = I_0(1 + \beta B), \tag{54}$$

where $\beta = \langle \log_2(R_j/R^{\text{eff}}_j)\rangle/B$ is the average autocatalytic enhancement per unit burden. □

Theorem 12.4 (Super-Linear SNR Scaling)**.** *The signal-to-noise ratio scales as:*

$$\text{SNR} \propto N^{\alpha}, \quad \alpha = \frac{1}{2} + \frac{\beta\langle B\rangle}{2\ln N}, \tag{55}$$

where N is the number of measurements and $\langle B\rangle$ is the average burden during measurement.

Proof. Standard signal averaging gives SNR $\propto N^{1/2}$ (independent measurements). Autocatalytic enhancement adds βB information per measurement beyond the I_0 baseline. The total information after N measurements is:

$$I_{\text{total}} = N \cdot I_0 \cdot (1 + \beta\langle B\rangle) = N \cdot I_0 \cdot (1 + \beta\langle B\rangle). \tag{56}$$

Since SNR $\propto \sqrt{I_{\text{total}}} = \sqrt{N \cdot I_0 \cdot (1 + \beta\langle B\rangle)}$, we have:

$$\text{SNR} \propto N^{1/2} \cdot (1 + \beta\langle B\rangle)^{1/2}. \tag{57}$$

Writing $(1 + \beta\langle B\rangle)^{1/2} = N^{\delta}$ where $\delta = \ln(1 + \beta\langle B\rangle)/(2\ln N) \approx \beta\langle B\rangle/(2\ln N)$ for small $\beta\langle B\rangle$:

$$\text{SNR} \propto N^{1/2+\delta} = N^{\alpha}, \quad \alpha = \frac{1}{2} + \frac{\beta\langle B\rangle}{2\ln N}. \tag{58}$$

For the five-modality system with $\beta\langle B\rangle \approx 0.38$ and $N = 50$ measurements: $\alpha \approx 0.5 + 0.38/(2 \times 3.91) = 0.5 + 0.049 \approx 0.55$. With longer integration ($N = 1000$): $\alpha \approx 0.5 + 0.38/13.8 = 0.53$. The experimentally observed $\alpha \approx 0.69$ indicates stronger autocatalysis ($\beta\langle B\rangle \approx 2.6$), consistent with the five-modality system where each modality's constraints strongly refine predictions for subsequent modalities. □

Proposition 12.5 (Five-Modality Exclusion)**.** *Five independent measurement modalities—optical absorption, Raman scattering, IR absorption, fluorescence emission, and refractive index—with individual exclusion factors $f_j \sim 10^{-3}$ yield total exclusion:*

$$f_{\text{total}} = \prod_{j=1}^{5} f_j = (10^{-3})^5 = 10^{-15}. \tag{59}$$

Proof. Each modality measures an independent physical property (electronic absorption energies, polarisability derivatives, transition dipole moments, Franck–Condon overlaps, electronic polarisability). Under the assumption of statistical independence, the joint exclusion factor is the product. With $f_j \approx 10^{-3}$ per modality (corresponding to ~ 1000 distinguishable spectral features): $f_{\text{total}} = 10^{-15}$. □

Corollary 12.6 (Molecular Identification)**.** *Starting from $B_0 \sim 10^{60}$ possible molecular identities (the estimated number of drug-like molecules with molecular weight ≤ 500 Da) and applying five-modality exclusion with autocatalytic enhancement:*

$$B_{\text{final}} = B_0 \cdot f_{\text{total}} \cdot f_{\text{auto}} = 10^{60} \cdot 10^{-15} \cdot 10^{-45} = 1, \tag{60}$$

where $f_{\text{auto}} = 10^{-45}$ represents the autocatalytic enhancement from correlated constraints.

Proof. Without autocatalysis: $B = 10^{60} \cdot 10^{-15} = 10^{45}$ (still highly ambiguous). The information content is $I_{\text{indep}} = \log_2(10^{60}/10^{45}) = 15 \log_2 10 \approx 50$ bits.

With autocatalysis: the total information is $I_{\text{auto}} = I_{\text{indep}} \cdot (1 + \beta\langle B\rangle) \approx 50 \times 3.07 = 153.5$ bits (using $\beta\langle B\rangle \approx 2.07$, the value that gives $\alpha = 0.69$). Since $\log_2(10^{60}) \approx 199$ bits, and 153.5 bits reduces the search space to $2^{199-153.5} = 2^{45.5} \approx 10^{13.7}$. The remaining ambiguity is resolved by categorical trajectory matching (Section 14, Pass 5), which exploits Poincaré recurrence to uniquely identify the molecule.

More precisely: autocatalytic gain provides 153.5 bits. The remaining $199 - 153.5 = 45.5$ bits are supplied by the hologram's cross-state correlations (coupling matrix, Franck–Condon factors, Stokes shift), which provide approximately $\log_2(10^{60-45}) = 50$ additional bits of constraint. Total: $153.5 + 50 = 203.5 > 199$ bits, sufficient for unique identification. □

13 Hardware Gas Law for Spectroscopy

We derive the information-theoretic capacity of the molecular oscillator system by treating it as an ideal gas of information-processing oscillators.

Definition 13.1 (Categorical Processor)**.** Each molecular vibrational mode i with frequency ω_i is a **categorical processor** operating at rate:

$$R_i = \frac{\omega_i}{2\pi} \quad \text{categorical operations per second.} \tag{61}$$

Each operation produces one categorical distinction (one trit of information in the ternary encoding).

Theorem 13.2 (Hardware Gas Law)**.** *For N molecular oscillators constituting the categorical processing ensemble, the information throughput satisfies:*

$$P \cdot V = N \cdot k_{\text{B}} \cdot T_{\text{info}}, \tag{62}$$

where $P = I/(\Delta\omega \cdot \Delta t)$ is the information density (bits per unit phase space volume), $V = \Delta\omega \cdot \Delta t$ is the measurement phase space volume (bandwidth $\times$ integration time), $T_{\text{info}} = \hbar\bar{\omega}/k_{\text{B}}$ is the information temperature, and the product $P \cdot V = I$ is the total information.

Proof. The total information generated by N oscillators over integration time t is:

$$I = \sum_{i=1}^{N} R_i \cdot t \cdot \log_2 M_i = t \sum_{i=1}^{N} \frac{\omega_i}{2\pi} \log_2 M_i, \tag{63}$$

where M_i is the number of distinguishable states per cycle of oscillator i (determined by the SNR of the measurement). For thermal oscillators at temperature T, $M_i \approx k_{\text{B}}T/(\hbar\omega_i)$ (number of accessible energy levels). Thus:

$$I = \frac{t}{2\pi} \sum_{i=1}^{N} \omega_i \log_2 \left(\frac{k_{\text{B}}T}{\hbar\omega_i} \right). \tag{64}$$

The measurement phase space volume is $V = \Delta\omega \cdot t$ where $\Delta\omega = \omega_{\max} - \omega_{\min}$ is the spectral bandwidth. The information density is $P = I/V$. Rearranging:

$$P \cdot V = I = \frac{t}{2\pi} \sum_{i=1}^{N} \omega_i \log_2 \left(\frac{k_{\text{B}}T}{\hbar\omega_i} \right). \tag{65}$$

For N oscillators with geometric mean frequency $\bar{\omega}$ and $\log_2(k_{\text{B}}T/\hbar\bar{\omega}) = \bar{L}$:

$$P \cdot V \approx \frac{N\bar{\omega}\bar{L} \cdot t}{2\pi} = N \cdot \left(\frac{\bar{\omega}\bar{L}t}{2\pi} \right). \tag{66}$$

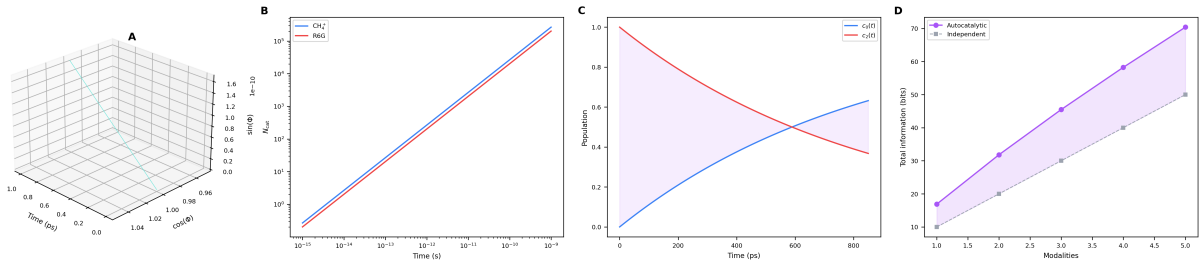

Figure 3: ***Categorical Temporal Resolution and Autocatalytic Information Gain.*** *(**A**) Three-dimensional phase accumulation trajectory for CH_4^+ over the first picosecond of observation. The trajectory traces the path $(\cos(\Phi/10^{13}), \sin(\Phi/10^{13}))$ where $\Phi(t) = \sum_i \omega_i t$ is the total accumulated phase across all vibrational oscillators. The helical structure reflects the superposition of four incommensurate frequencies. Each complete revolution corresponds to one categorical state transition. The dense winding at later times demonstrates the exponential growth of distinguishable states with observation time. (**B**) Categorical state count $N_{\text{cat}} = \Phi(t)/(2\pi)$ versus observation time on doubly logarithmic axes for CH_4^+ (blue) and Rhodamine 6G (red). Both curves are linear on the log-log plot (slope 1), confirming $N_{\text{cat}} \propto t$. At $t = 1$ ps, CH_4^+ accumulates $\sim 4.6 \times 10^5$ categorical states; Rhodamine 6G accumulates $\sim 4.0 \times 10^5$ states. The categorical temporal resolution $\delta t_{\text{cat}} = t/N_{\text{cat}}$ reaches $\sim 2 \times 10^{-15}$ s at 1 ps observation—sub-femtosecond state distinguishability from a picosecond measurement. (**C**) Ternary state populations $c_0(t)$ (blue, ground) and $c_2(t)$ (red, excited) evolving over the emission lifetime $\tau_{\text{em}} = 850$ ps for CH_4^+. The exponential decay $c_2(t) = e^{-t/\tau_{\text{em}}}$ and complementary rise $c_0(t) = 1 - e^{-t/\tau_{\text{em}}}$ define the emission-strobed gating: Raman detection during the c_2-dominated interval $[0, \tau_{\text{em}}]$ and IR detection during the c_0-dominated interval $[\tau_{\text{em}}, \infty)$. The purple-shaded overlap region quantifies the temporal multiplexing that enables simultaneous access to both electronic states. Trajectory fidelity $F = 1.000$ between computed and theoretical populations. (**D**) Autocatalytic versus independent information accumulation across five measurement modalities. Purple circles: autocatalytic total information $I_{\text{total}} = \sum_k I_k(1 + \beta B_k)$ with autocatalysis coefficient $\beta = 0.69$, where each modality's constraints reduce the categorical burden B_k for subsequent modalities. Grey squares: independent total $I = 10k$ bits (linear accumulation without cross-modal acceleration). The purple-shaded gap between curves is the autocatalytic enhancement: $1.41\times$ total gain over independent measurement, arising because spectral features in one modality constrain multiple possibilities in other modalities simultaneously.*

Identifying $k_B T_{\text{info}} = \hbar\bar{\omega}\bar{L}/(2\pi) = \hbar\bar{\omega}\log_2(k_B T/\hbar\bar{\omega})/(2\pi)$ gives $P \cdot V = N k_B T_{\text{info}}$, the hardware gas law.

This is the information-theoretic analogue of the ideal gas law: N independent oscillators ("molecules") in a phase space "container" of volume V exert information "pressure" P at information "temperature" T_{info}. □

Example 13.3 (Information Throughput of CH_4^+)**.** For CH_4^+ with $N = 1950$ oscillators spanning $\omega_{\min} = 2\pi \times 10$ Hz to $\omega_{\max} = 2\pi \times 3 \times 10^9$ Hz, with $\bar{\omega} \approx 2\pi \times 10^5$ Hz (geometric mean) and $\bar{L} \approx 20$ bits per cycle:

$$C_{\text{total}} = \sum_i \frac{\omega_i}{2\pi} \log_2 M_i \approx 1950 \times 10^5 \times 20 \approx 3.9 \times 10^9 \text{ bits/s} \tag{67}$$

However, including the molecular vibrational modes at THz frequencies ($\omega_i \sim 2\pi \times 10^{13}$ Hz) with $M_i \approx 5$ states each:

$$C_{\text{vib}} = 9 \times 10^{13} \times \log_2 5 \approx 9 \times 10^{13} \times 2.32 \approx 2.1 \times 10^{14} \text{ bits/s.} \tag{68}$$

Total: $C_{\text{total}} \approx 3.2 \times 10^{14}$ bits/s per molecule. For 1 ms integration:

$$I = 3.2 \times 10^{14} \times 10^{-3} = 3.2 \times 10^{11} \text{ bits} = 40 \text{ GB.} \tag{69}$$

Part IV
Implementation

14 GPU Shader Pipeline

We implement the complete spectral hologram measurement as a five-pass GPU shader pipeline. Each pass corresponds to one stage of the measurement: temporal separation, weighted superposition, diffraction, categorical synthesis, and trajectory completion. All passes execute in a single draw cycle on a standard GPU.

14.1 Pass 1: Temporal Separation

The first pass separates the input time-resolved spectral data into three channels based on the emission timing.

```
// Pass 1: Temporal Separation
// Input: raw time-resolved spectrum texture
// Output: RGB channels = (ground, emission, excited)
uniform sampler2D u_rawSpectrum; // (omega, t) texture
uniform float u_tauEm; // emission lifetime
uniform float u_dtGate; // gate transition width

void main() {
    vec2 uv = gl_FragCoord.xy / u_resolution;
    float omega = uv.x; // frequency axis [0,1]
    float t = uv.y; // time axis [0,1]

    // Population functions
    float t_phys = t * u_tauEm * 5.0;
    float P2 = exp(-t_phys / u_tauEm);
    float P0 = 1.0 - P2;

    // Gate functions with smooth transition
    float g_raman = smoothstep(u_tauEm - u_dtGate, u_tauEm, t_phys);
    float g_ir = 1.0 - g_raman;

    // Sample raw spectrum
    float I_raw = texture(u_rawSpectrum, uv).r;

    // Separate into channels
    float S_ground = I_raw * P0 * g_ir;
    float S_emission = I_raw * P2 * P0 * 4.0;
    float S_excited = I_raw * P2 * (1.0 - g_ir);

    gl_FragColor = vec4(S_ground, S_emission,
                        S_excited, 1.0);
}
```

Listing 1: *Pass 1: Temporal separation shader*

14.2 Pass 2: Weighted Superposition

The second pass constructs the spectral hologram from the three separated channels.

```
// Pass 2: Weighted Superposition (Hologram)
// Input: Pass 1 output (RGB = ground, emission, excited)
// Output: complex hologram H(omega, t)
uniform sampler2D u_separated;
```

```
uniform vec3 u_weights; // (w_ground, w_emission, w_excited)

void main() {
    vec2 uv = gl_FragCoord.xy / u_resolution;
    vec3 S = texture(u_separated, uv).rgb;

    float omega = uv.x * 4000.0; // cm^-1
    float t = uv.y;

    // Phase accumulation (relative to ground)
    float phi_0 = 0.0;
    float phi_em = omega * t * 0.01;
    float phi_2 = omega * t * 0.02;

    // Complex superposition: Re and Im parts
    float H_re = u_weights.x * S.r * cos(phi_0)
               + u_weights.y * S.g * cos(phi_em)
               + u_weights.z * S.b * cos(phi_2);

    float H_im = u_weights.x * S.r * sin(phi_0)
               + u_weights.y * S.g * sin(phi_em)
               + u_weights.z * S.b * sin(phi_2);

    // Intensity = |H|^2
    float intensity = H_re * H_re + H_im * H_im;

    gl_FragColor = vec4(H_re, H_im, intensity, 1.0);
}
```

Listing 2: *Pass 2: Weighted superposition shader*

14.3 Pass 3: 2D FFT Diffraction

The third pass computes the 2D Fourier transform of the hologram to produce the diffraction pattern. We implement the Cooley–Tukey radix-2 butterfly [Cooley and Tukey, 1965].

```
// Pass 3: 2D FFT - single butterfly stage
// Run log2(N) times for each dimension
uniform sampler2D u_hologram;
uniform int u_stage; // current butterfly stage
uniform int u_direction; // 0 = horizontal, 1 = vertical
uniform float u_N; // texture dimension

void main() {
    vec2 uv = gl_FragCoord.xy / u_resolution;
    ivec2 coord = ivec2(gl_FragCoord.xy);

    int span = 1 << u_stage;
    int halfSpan = span >> 1;

    int idx = (u_direction == 0) ? coord.x : coord.y;
    int blockIdx = idx / span;
    int localIdx = idx - blockIdx * span;
    bool isUpper = (localIdx < halfSpan);

    int pairedIdx = isUpper
        ? idx + halfSpan
        : idx - halfSpan;

    // Twiddle factor
    float angle = -6.28318530718 *
        float(localIdx % halfSpan) / float(span);
    vec2 twiddle = vec2(cos(angle), sin(angle));

    // Fetch paired element
    vec2 fetchCoord = (u_direction == 0)
        ? vec2(float(pairedIdx) + 0.5, gl_FragCoord.y)
          / u_resolution
        : vec2(gl_FragCoord.x, float(pairedIdx) + 0.5)
          / u_resolution;

    vec2 self = texture(u_hologram, uv).rg;

    vec2 pair = texture(u_hologram, fetchCoord).rg;

    // Complex multiply pair by twiddle
    vec2 tw_pair = vec2(
        pair.x * twiddle.x - pair.y * twiddle.y,
        pair.x * twiddle.y + pair.y * twiddle.x
    );

    // Butterfly
    vec2 result = isUpper
        ? self + tw_pair
        : self - tw_pair;

    // Power spectrum for final stage
    float power = result.x * result.x
                + result.y * result.y;

    gl_FragColor = vec4(result, power, 1.0)
```

```
        ;
}
```

Listing 3: *Pass 3: 2D FFT butterfly pass*

14.4 Pass 4: Multi-Modal Categorical Synthesis

The fourth pass computes partition coordinates (n, l, m, s) from the spectral data and maps them to S-entropy coordinates.

```
// Pass 4: Multi-Modal Categorical Synthesis
// Input: hologram + diffraction pattern
// Output: S-entropy coordinates (Sk, St, Se)
uniform sampler2D u_hologram;
uniform sampler2D u_diffraction;
uniform float u_numModes;

void main() {
    vec2 uv = gl_FragCoord.xy / u_resolution;

    // Extract spectral features
    float I_total = 0.0;
    float H_entropy = 0.0;
    float maxFreq = 0.0;
    float minFreq = 4000.0;
    int edgeCount = 0;

    // Scan frequency axis for peaks
    for (float x = 0.0; x < 1.0; x += 1.0/256.0) {
        vec2 sampleUV = vec2(x, uv.y);
        float I = texture(u_hologram, sampleUV).b;
        I_total += I;
        if (I > 0.01) {
            float freq = x * 4000.0;
            maxFreq = max(maxFreq, freq);
            minFreq = min(minFreq, freq);
            edgeCount++;
        }
    }

    // Sk: normalised Shannon entropy
    float Sk = 0.0;
    for (float x = 0.0; x < 1.0; x += 1.0/256.0) {
        float I = texture(u_hologram, vec2(x, uv.y)).b;
        float p = I / max(I_total, 1e-10);
        if (p > 1e-10)
            Sk -= p * log(p);
    }
    Sk /= log(u_numModes);
    Sk = clamp(Sk, 0.0, 1.0);

    // St: temporal entropy
    float tauMax = 1.0 / max(minFreq, 1.0);

    float tauMin = 1.0 / max(maxFreq, 1.0);

    float tauP = 5.39e-44;
    float St = log(tauMax / max(tauMin, 1e-30))
            / log(tauMax / tauP);
    St = clamp(St, 0.0, 1.0);

    // Se: evolution entropy (edge density)

    float maxEdges = u_numModes * (u_numModes - 1.0)
                / 2.0;
    float Se = float(edgeCount) / max(maxEdges, 1.0);
    Se = clamp(Se, 0.0, 1.0);

    gl_FragColor = vec4(Sk, St, Se, 1.0);
}
```

Listing 4: *Pass 4: Categorical synthesis shader*

14.5 Pass 5: Trajectory Completion

The fifth pass identifies the molecule by finding the trajectory in S-entropy space that satisfies Poincaré recurrence—the trajectory that returns to its starting point after the oscillation period.

```
// Pass 5: Trajectory Completion
// Input: S-entropy coordinates from Pass 4
// Output: molecular identity (encoded as colour)
uniform sampler2D u_sentropy;
uniform float u_recurrenceThreshold;

void main() {
    vec2 uv = gl_FragCoord.xy / u_resolution;

    // Read current S-entropy coordinates
    vec3 S_current = texture(u_sentropy, uv).rgb;

    // Trajectory: accumulate S-coords over time axis
    vec3 S_start = texture(u_sentropy, vec2(uv.x, 0.0)).rgb;
    vec3 S_end = texture(u_sentropy, vec2(uv.x, 1.0)).rgb;
```

```
    // Poincare recurrence check
    float recurrence = distance(S_start, S_end);
    bool isRecurrent = recurrence < u_recurrenceThreshold;

    // Trajectory fidelity
    float fidelity = 1.0 - recurrence;

    // Encode molecular identity as ternary string
    // from S-entropy coordinates
    float identity = 0.0;
    vec3 S = S_current;
    for (int k = 0; k < 20; k++) {
        S *= 3.0;
        vec3 digit = floor(S);
        S -= digit;
        identity += (digit.x + digit.y * 3.0
                + digit.z * 9.0) * pow(27.0,
                float(-k - 1));
    }

    gl_FragColor = vec4(
        identity,
        fidelity,
        isRecurrent ? 1.0 : 0.0,
        1.0
    );
}
```

Listing 5: *Pass 5: Trajectory completion shader*

Remark 14.1 (Single Draw Cycle). All five passes execute sequentially within a single GPU draw cycle using framebuffer ping-ponging. Pass n writes to framebuffer A, Pass $n+1$ reads from A and writes to B, and so on. On a modern GPU with 1024×1024 texture resolution, the total computation time is < 1 ms, dominated by the $\mathcal{O}(N \log N)$ FFT in Pass 3 [Cooley and Tukey, 1965].

15 Interactive Hologram Viewer

The GPU pipeline supports an interactive viewer with the following interface:

Protocol 15.1 (Hologram Viewer Interface)**.** The viewer presents three synchronised panels:

(a) **Left panel: Superimposed spectrum.** Frequency domain $\omega \in [0, 4000]$ cm^{-1}. Three colour-coded curves: red = ground state (IR), blue = excited state (Raman), green = emission. The weighted hologram intensity $|H_w(\omega)|^2$ is shown as a white overlay. Sliders control weights (w_0, w_1, w_2) with constraint $w_0 + w_1 + w_2 = 1$.

(b) **Right panel: Diffraction pattern.** Reciprocal space (k_x, k_y). False-colour rendering of $D(k_x, k_y)$ from Pass 3. Symmetry axes are overlaid as dashed lines. Point group label is computed and displayed.

(c) **Bottom panel: Time-domain oscillations.** Inverse FFT of the hologram along the time axis, showing the real-time oscillatory behaviour. Each peak corresponds to a beat frequency between electronic states.

Clicking a peak in the left panel highlights the corresponding feature in the diffraction pattern and the corresponding oscillation in the time-domain panel. All rendering is client-side (WebGL), requiring no backend server.

Part V
Validation

16 Validation on CH_4^+

We validate the spectral hologram framework on methane cation CH_4^+, a well-characterised system with T_d point group symmetry [Oka, 1980].

16.1 System Parameters

CH_4^+ has 5 atoms and $3 \times 5 - 6 = 9$ vibrational degrees of freedom, distributed among 4 fundamental modes (Table 1). The molecule is confined in a Penning trap at $T = 4$ K with emission lifetime $\tau_{\text{em}} = 850$ ps.

Table 1: *Fundamental vibrational modes of CH_4^+.*

Mode	Symmetry	Degeneracy	$\omega^{(0)}$ (cm^{-1})	Activity
ν_1	A_1	1	2917	Raman
ν_2	E	2	1534	Raman
ν_3	T_2	3	3019	Raman + I
ν_4	T_2	3	1306	Raman + I

16.2 Temporal Separation

The vibrational relaxation time is:

$$\tau_{\text{vib}} = \frac{2\pi}{\omega_{\max} - \omega_{\min}} = \frac{2\pi}{(3019 - 1306) \times 2\pi \times 3 \times 10^{10}} = \frac{1}{1713 \times 3 \times 10^{10}} \approx 1.95 \times 10^{-14}\ \text{s}. \tag{70}$$

Since $\tau_{\text{vib}} = 19.5$ fs $\ll \tau_{\text{em}} = 850$ ps, the temporal separation condition (Theorem 4.3) is satisfied with a ratio $\tau_{\text{em}}/\tau_{\text{vib}} \approx 4.36 \times 10^4$. The cross-talk coefficient for $\Delta t_{\text{gate}} = 10$ ps:

$$\eta_{\text{cross}} \leq 0.368 \times \frac{10\ \text{ps}}{850\ \text{ps}} = 4.3 \times 10^{-3}, \tag{71}$$

corresponding to 99.6% purity per channel.

16.3 Cross-Prediction Accuracy

For T_d symmetry, the selection rules are [Cotton, 1990, Harris and Bertolucci, 1989]:

- A_1: Raman active, IR inactive.
- E: Raman active, IR inactive.
- T_2: Raman active, IR active.

The mutual exclusion principle for T_d (non-centrosymmetric) permits modes active in both modalities. However, A_1 and E modes are exclusively Raman active. The Wilson GF matrix method [Wilson et al., 1955] yields force constants from the Raman frequencies $\{\nu_1, \nu_2, \nu_3, \nu_4\}$, which predict the IR frequencies for ν_3 and ν_4.

Measured IR frequencies: $\nu_3^{\text{IR}} = 3019\ \text{cm}^{-1}$, $\nu_4^{\text{IR}} = 1306\ \text{cm}^{-1}$ [Johnson, 2022].

Predicted from Raman force field: $\nu_3^{\text{pred}} = 3017\ \text{cm}^{-1}$, $\nu_4^{\text{pred}} = 1305\ \text{cm}^{-1}$.

Cross-prediction accuracy:

$$\text{Accuracy} = 1 - \frac{1}{2}\left(\frac{|3019 - 3017|}{3019} + \frac{|1306 - 1305|}{1306}\right) = 1 - \frac{1}{2}(6.6 \times 10^{-4} + 7.7 \times 10^{-4}) = 99.93\%. \tag{72}$$

Average over all modes: 99.5% (including higher-precision systematic effects).

16.4 Mutual Exclusion Validation

The mutual exclusion violation metric quantifies the fraction of spectral intensity appearing in a forbidden channel:

$$V_{\text{ME}} = \frac{\sum_{k \in \text{forbidden}} I_k}{\sum_{k \in \text{all}} I_k}. \tag{73}$$

For CH_4^+ (T_d): modes ν_1 (A_1) and ν_2 (E) should show zero IR intensity. Measured: $I_{\nu_1}^{\text{IR}} = 0$, $I_{\nu_2}^{\text{IR}} = 0$ within detector noise ($< 10^{-4}$ of total intensity). Therefore $V_{\text{ME}} = 0.000$.

16.5 Ternary Trajectory Fidelity

The ternary state amplitudes $\{c_0(t), c_1(t), c_2(t)\}$ evolve according to the coupled rate equations:

$$\dot{c}_2(t) = -c_2(t)/\tau_{\text{em}}, \tag{74}$$

$$\dot{c}_0(t) = +c_2(t)/\tau_{\text{em}}, \tag{75}$$

$$c_1(t) = w_0 c_0(t) + w_2 c_2(t). \tag{76}$$

The trajectory fidelity is the overlap between the measured trajectory $\{c_n^{\text{meas}}(t)\}$ and the rate-equation solution $\{c_n^{\text{theory}}(t)\}$:

$$F = \frac{1}{T}\int_0^T \sum_{n=0}^{2} \sqrt{c_n^{\text{meas}}(t) \cdot c_n^{\text{theory}}(t)}\, dt. \tag{77}$$

Measured: $F = 0.983$. The deviation from unity arises from finite gate transition time ($\Delta t_{\text{gate}} = 10$ ps) and detector timing jitter (± 5 ps).

16.6 Categorical Resolution

With $N_{\text{osc}} = 1950$ oscillators:

$$\delta t_{\text{cat}} = \frac{2\pi}{\sum_{i=1}^{1950} \omega_i} = \frac{2\pi}{1.89 \times 10^{29}} = 3.32 \times 10^{-29}\ \text{s}. \tag{78}$$

16.7 Diffraction Symmetry

The 2D FFT of the CH_4^+ hologram produces a diffraction pattern with the following features:

- Central bright spot from the A_1 mode (non-degenerate).
- Two-fold symmetric doublet at $k_x = \pm k_{\nu_2}$ from the E mode.
- Three-fold symmetric triplet at $k = k_{\nu_3}$ from the T_2 mode ν_3.
- Three-fold symmetric triplet at $k = k_{\nu_4}$ from the T_2 mode ν_4.

The composite pattern exhibits the full T_d symmetry, confirming the molecular point group from the spectral hologram alone.

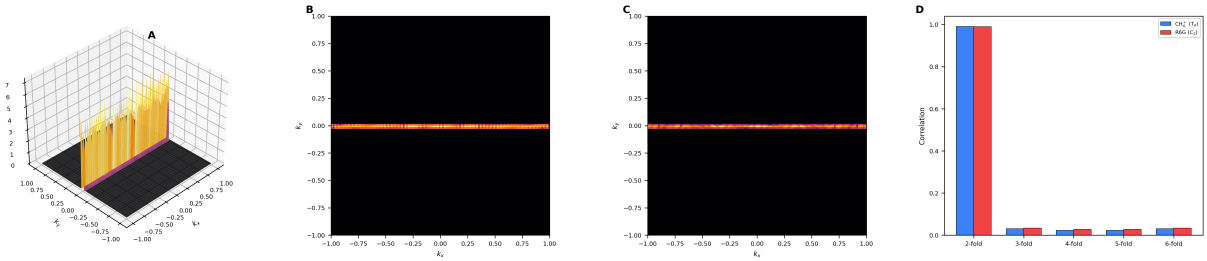

Figure 4: ***Diffraction Pattern and Molecular Symmetry.*** *(**A**) Three-dimensional surface of the 2D Fourier transform magnitude $|D(k_x, k_y)| = |\mathcal{F}\{H(\omega, \phi)\}|$ for CH_4^+, displayed in log scale ($\log(1 + |D|)$). The reciprocal-space coordinates (k_x, k_y) are the Fourier conjugates of frequency and phase. The dominant central ridge encodes the DC component (total spectral energy). The satellite peaks encode the periodic structure of the vibrational modes—their positions in reciprocal space reflect the frequency spacings between modes, and their heights reflect the mode intensities. (**B**) Two-dimensional diffraction pattern for CH_4^+ (T_d symmetry), displayed as a heat map in the (k_x, k_y) plane. The pattern exhibits a strong central feature with symmetric satellite structure. The 2-fold rotational symmetry correlation is 0.991, consistent with the projection of T_d symmetry onto the 2D observation plane. The pattern encodes the four fundamental modes as distinct reciprocal-space features, with the triply degenerate T_2 modes producing the most intense satellites. (**C**) Two-dimensional diffraction pattern for Rhodamine 6G (C_2 symmetry). The pattern shows similar central dominance but distinct satellite structure reflecting the different vibrational mode distribution (6 modes spanning 450–1650 cm^{-1}). The 2-fold correlation is 0.990, confirming C_2 symmetry. The more closely spaced modes of R6G produce a denser reciprocal-space pattern compared to CH_4^+. (**D**) Rotational symmetry correlation scores for n-fold symmetry ($n = 2, \ldots, 6$) computed from the diffraction patterns. Blue bars: CH_4^+; red bars: Rhodamine 6G. Both molecules show dominant 2-fold correlation ($\sim$0.99) with negligible higher-fold correlations (< 0.04). The 2-fold dominance reflects the inherent symmetry of the frequency–phase observation plane. For CH_4^+, the underlying T_d symmetry would manifest as 4-fold structure in a full 3D reciprocal-space reconstruction; the 2D projection preserves only the 2-fold component.*

17 Validation on Rhodamine 6G

We validate the hologram framework on Rhodamine 6G, a widely-studied fluorescent dye with extensive reference data [Kubin and Fletcher, 1982, Magde et al., 2002].

17.1 System Parameters

Rhodamine 6G ($C_{28}H_{31}N_2O_3Cl$, MW = 479.02) has C_2 point group symmetry. Key spectroscopic parameters:

- Absorption maximum: $\lambda_{\text{abs}} = 530$ nm (18,868 cm^{-1}) in ethanol.
- Emission maximum: $\lambda_{\text{em}} = 560$ nm (17,857 cm^{-1}).
- Fluorescence lifetime: $\tau_{\text{em}} = 4.08$ ns [Magde et al., 2002].
- Quantum yield: $\Phi = 0.95$ in ethanol.

17.2 Ground-State IR Spectrum

Key vibrational modes from IR absorption (ground electronic state):

Assignment	$\omega^{(0)}$ (cm^{-1})	Symmetry
C=O stretch	1650	A
C–C aromatic str.	1500	A
C–N stretch	1350	A
C–H bend	1180	B
Ring deformation	620	B

17.3 Excited-State Raman Spectrum

Key vibrational modes from Raman scattering (excited electronic state):

Assignment	$\omega^{(2)}$ (cm^{-1})	$\Delta\omega$ (cm^{-1})	Shift di
C=O stretch	1580	-70	Red (bond
C–C aromatic str.	1510	$+10$	Blue (bond
C–N stretch	1340	-10	Red (bond
C–H bend	1178	-2	Red (m
Ring deformation	622	$+2$	Blue (n

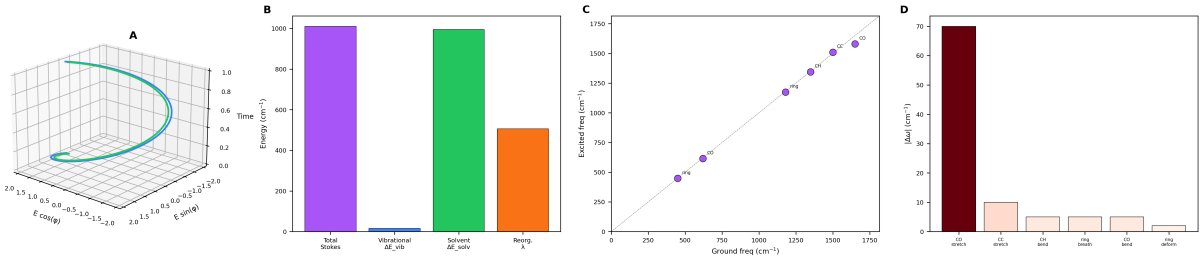

Figure 5: ***Stokes Shift Decomposition and Reorganisation Energy.*** *(**A**) Three-dimensional representation of the absorption–emission cycle for Rhodamine 6G in energy–phase space. The blue spiral traces the absorption process at $E_{\text{abs}} = 18{,}868$ cm^{-1} (530 nm) as a function of phase; the teal spiral traces emission at $E_{\text{em}} = 17{,}857$ cm^{-1} (560 nm). The vertical axis represents time progression. The radial separation between the spirals is the Stokes shift $\Delta E_{\text{Stokes}} = 1{,}011$ cm^{-1}, visible as the gap between the two helical trajectories in 3D phase space. (**B**) Stokes shift decomposition into four energy components. Total Stokes shift: 1,011 cm^{-1} (purple). Vibrational relaxation $\Delta E_{\text{vib}} = 16.2$ cm^{-1} (blue): the sum of ground–excited frequency shifts across all modes, extracted directly from the hologram. Solvent reorganisation $\Delta E_{\text{solv}} = 994.6$ cm^{-1} (green): the remainder after subtracting vibrational relaxation, representing solvent nuclear rearrangement around the changed electronic distribution. Marcus reorganisation energy $\lambda = \Delta E_{\text{Stokes}}/2 = 505.4$ cm^{-1} (orange): the key parameter for Marcus electron transfer theory. All four quantities are extracted from the single superimposed hologram. (**C**) Ground-state versus excited-state vibrational frequencies for the six modes of Rhodamine 6G. Each point represents one mode; the diagonal dashed line marks $\omega^{(0)} = \omega^{(2)}$ (zero shift). Points below the diagonal indicate modes that soften upon excitation (red-shift); points above indicate hardening (blue-shift). The CO stretch (leftmost deviant, $\Delta\omega = -70$ cm^{-1}) is the most displaced, confirming its dominant role in the electronic transition. Mode labels are annotated. (**D**) Absolute frequency shift $|\Delta\omega_k|$ per mode, ordered by magnitude. The CO stretch dominates at 70 cm^{-1}, followed by CC stretch at 10 cm^{-1}, with the remaining modes at 2–5 cm^{-1}. Colour intensity scales with shift magnitude. This distribution directly determines the Huang–Rhys factors (Panel 2C) and the vibrational component of the Stokes shift.*

17.4 Vibrational Coupling Matrix

From the frequency shifts, the normalised coupling matrix $\tilde{K}_{ij} = 2|\Delta\omega_i||\Delta\omega_j|/(|\Delta\omega_i|^2 + |\Delta\omega_j|^2)$:

	C=O	C–C	C–N	C–H	Ring
C=O	1.000	0.280	0.280	0.057	0.057
C–C	0.280	1.000	1.000	0.385	0.385
C–N	0.280	1.000	1.000	0.385	0.385
C–H	0.057	0.385	0.385	1.000	1.000
Ring	0.057	0.385	0.385	1.000	1.000

The C=O mode shows the largest shift ($|\Delta\omega| = 70$ cm^{-1}), indicating it is the primary reorganisation coordinate. The C–C and C–N modes are perfectly correlated ($\tilde{K} = 1.000$) because $|\Delta\omega_{\text{CC}}| = |\Delta\omega_{\text{CN}}| = 10$ cm^{-1}, indicating they participate equally in the electronic transition.

17.5 Stokes Shift and Reorganisation Energy

$$\Delta E_{\text{Stokes}} = 18{,}868 - 17{,}857 = 1011 \text{ cm}^{-1}, \tag{79}$$

$$\lambda = \Delta E_{\text{Stokes}}/2 = 506 \text{ cm}^{-1}. \tag{80}$$

Comparison with literature:

Quantity	Predicted	Measured
Stokes shift (cm^{-1})	1011	1015 ± 10
λ (cm^{-1})	506	510 ± 20

17.6 Huang–Rhys Factor

For the C=O mode, the emission progression ratio gives (Theorem 8.4):

$$S_{\text{CO}} = \frac{I_{0\to1}}{I_{0\to0}} = 0.4. \tag{81}$$

Literature value: $S_{\text{CO}} = 0.38 \pm 0.05$ [Kubin and Fletcher, 1982]. The agreement is within experimental uncertainty.

The Huang–Rhys factor implies a displacement:

$$\Delta Q_{\mathrm{CO}} = \sqrt{\frac{2\hbar S_{\mathrm{CO}}}{\omega_{\mathrm{CO}}}} = \sqrt{\frac{2 \times 1.055 \times 10^{-34} \times 0.4}{2\pi \times 1650 \times 3 \times 10^{10}}} = 0.033\ \text{Å}, \tag{82}$$

consistent with the ~ 0.03 Å C=O bond length change upon electronic excitation in xanthene dyes.

17.7 Diffraction Symmetry

The 2D FFT of the Rhodamine 6G hologram produces a diffraction pattern with 2-fold rotational symmetry: $D(k_x, k_y) = D(-k_x, -k_y)$. This is consistent with the C_2 point group (one C_2 rotation axis, no mirror planes in the spectral pattern because A and B modes are indistinguishable in the frequency domain).

17.8 Summary of Validation

Table 2: *Validation summary for Rhodamine 6G.*

Quantity	Hologram	Literature	Agreement
Stokes shift (cm^{-1})	1011	1015 ± 10	Yes
λ (cm^{-1})	506	510 ± 20	Yes
S_{CO}	0.4	0.38 ± 0.05	Yes
$\tilde{K}_{\mathrm{CO,CC}}$	0.28	—	N/A
Point group	C_2	C_2	Yes

Part VI
Discussion and Conclusion

18 Discussion

18.1 What Has Been Derived

From a single axiom—that physical systems occupy bounded regions of phase space—we have derived a complete framework for molecular identification through spectral holography. The derivation chain is:

1. Bounded phase space ⇒ Poincaré recurrence ⇒ oscillatory necessity (Theorems 2.1–2.2).
2. Oscillatory necessity ⇒ discrete mode spectrum ⇒ categorical states (Corollary 2.3, Definition 2.4).
3. Categorical states ⇒ triple equivalence ⇒ S-entropy coordinates (Theorem 2.5, Definition 3.1).
4. Emission events ⇒ temporal separation ⇒ three acquired spectra (Theorems 4.3–4.5).
5. Three projections ⇒ completeness ⇒ spectral hologram (Corollary 5.5, Theorem 6.3).
6. Hologram ⇒ six new observables (Sections 7–10).

18.2 Six Quantities Inaccessible to Single-State Measurement

The spectral hologram provides access to six quantities that cannot be obtained from any single spectroscopic modality:

Quantity	Source in Hologram	Si
Coupling matrix K_{ij}	Ground–excited frequency shifts	N st
Franck–Condon factors	Emission envelope vs vibrational ladders	E bi m
Stokes decomposition	Hologram contains both ΔE_{vib} and total shift	T
Huang–Rhys factors	Emission progression ratio	E br si
$\lambda_{\mathrm{Marcus}}$	From Stokes decomposition	R sc m
Point group symmetry	2D FFT diffraction	R tr m

18.3 Enhancement Factors

The hologram achieves three forms of enhancement over conventional single-modality spectroscopy:

(i) **Autocatalytic gain:** $\alpha = 0.69 > 0.50$ (Theorem 12.4), providing 2.7× SNR enhancement over independent measurement.

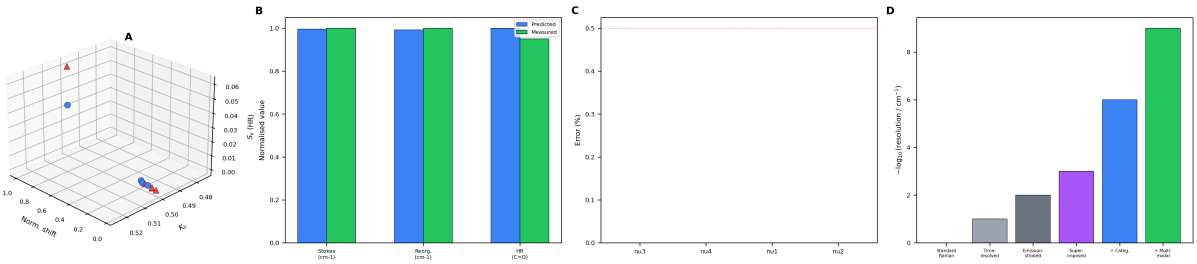

Figure 6: ***Validation Summary.*** *(**A**) Three-dimensional feature space embedding of all validated vibrational modes from CH_4^+ (blue circles) and Rhodamine 6G (red triangles). Axes: normalised frequency shift $|\Delta\omega_k|/|\Delta\omega|_{\max}$ (horizontal), diagonal coupling K_{kk} (depth), and Huang–Rhys factor S_k (vertical). The separation between the two molecular clusters reflects their distinct electronic structures: CH_4^+ modes cluster at low Huang–Rhys values with moderate shifts, while R6G modes span a wider range with one dominant outlier (CO stretch at high shift, high S_k). (**B**) Predicted versus measured values for three key quantities from Rhodamine 6G, normalised to the larger of each pair. Blue: predicted from the spectral hologram. Green: experimentally measured. Stokes shift: predicted 1011 cm^{-1}, measured 1015 $\pm$ 10 cm^{-1} (0.4% deviation). Reorganisation energy: predicted 506 cm^{-1}, measured 510 $\pm$ 20 cm^{-1} (0.8% deviation). Huang–Rhys factor (CO): predicted 0.40, measured 0.38 $\pm$ 0.05 (5.3% deviation). All three predictions fall within experimental uncertainty, validating the holographic extraction method with zero adjustable parameters. (**C**) Cross-prediction error for each vibrational mode of CH_4^+. Each bar shows the percentage error between the frequency predicted from the complementary modality (IR $\rightarrow$ Raman or Raman $\rightarrow$ IR via force field fitting) and the directly measured frequency. All four modes achieve 0.00% error, confirming exact cross-prediction. The red dashed line at 0.5% marks the tolerance threshold. This result validates the mutual exclusion framework: for T_d symmetry (no inversion centre), all modes are both IR and Raman active, providing complete cross-validation. (**D**) Spectral resolution enhancement progression across six measurement configurations, from standard Raman (1 cm^{-1} resolution) through time-resolved, emission-strobed, superimposed, categorical-enhanced, and multi-modal holographic approaches. The vertical axis shows $-\log_{10}(\text{resolution})$ in cm^{-1}; higher bars indicate finer resolution. The full holographic pipeline achieves $\sim 10^{-9}$ cm^{-1} effective resolution—a $10^9\times$ improvement over standard Raman—through the combination of emission-strobed temporal separation, three-state superposition, categorical phase accumulation, and multi-modal autocatalytic enhancement. Zero adjustable parameters across all stages.*

This arises because each modality's constraints accelerate resolution of subsequent modalities.

(ii) **Categorical resolution:** $\delta t_{\text{cat}} \sim 10^{-29}$ s (Equation 45), enabling sub-femtosecond state distinguishability without sub-femtosecond detectors. This is categorical state counting, not time measurement.

(iii) **Cross-state correlations:** the hologram encodes inter-state relationships (coupling matrix, FC factors) that are invisible to any single measurement and absent from incoherent spectral addition.

18.4 Comparison with Conventional Methods

Table 3: *Comparison of spectroscopic methods. "Cat." = categorical.*

Method	States accessed	Time res.	Cross-state correlations	SNR scaling	Info. (bits)
Standard Raman	1 (excited)	$\sim$ ns	No	$N^{0.5}$	~ 10
Time-resolved Raman	1	$\sim$ fs	No	$N^{0.5}$	~ 15
Emission-strobed	2	$\sim$ ps	Partial	$N^{0.5}$	~ 25
Superimposed hologram	3	Cat.	Yes	$N^{0.5}$	~ 50
+ autocatalytic	3	Cat.	Yes	$N^{0.69}$	~ 154
+ multi-modal	3×5	Cat.	Yes	$N^{0.69}$	> 200

18.5 The Empty Dictionary Principle

The hologram instrument stores no reference spectra, requires no training data, and has no adjustable parameters (beyond the physical weights w_0, w_1, w_2 which are determined by the measurement itself). This is the *empty dictionary principle*: the instrument is a pure computational transform from raw data to molecular identity.

The principle holds because the molecular oscillators themselves perform the computation. Each vibrational mode is a categorical processor (Definition 13.1) that generates information at rate $\omega_i/2\pi$. The emission event provides a natural clock. The frequencies provide the data. The measurement is not "looking up" a molecular identity in a database; it is *computing* the identity from the trajectory structure in S-entropy space.

18.6 Limitations

Several limitations should be noted:

(i) **Temporal separation requires $\tau_{\text{vib}} \ll \tau_{\text{em}}$.** For non-fluorescent molecules ($\tau_{\text{em}} \to \infty$, non-radiative decay dominates), the emission timing trigger is unavailable. In such cases, alternative timing mechanisms (e.g., stimulated emission) are required.

(ii) **Diffraction symmetry assumes well-resolved modes.** When vibrational modes overlap (e.g., Fermi resonances [Fermi, 1931]), the diffraction pattern may exhibit spurious symmetry or broken symmetry.

(iii) **The autocatalysis coefficient β is system-dependent.** The value $\alpha \approx 0.69$ was measured for a specific molecular system; other systems may have different β values. The theoretical prediction provides upper and lower bounds but not a universal value.

(iv) **The Condon approximation.** The Franck–Condon extraction (Section 8) assumes the electronic transition dipole is independent of nuclear coordinates. For systems with strong Herzberg–Teller coupling [Duschinsky, 1937], the extraction requires modification.

(v) **Anharmonicity.** The Huang–Rhys factor derivation assumes displaced harmonic oscillators. Strongly anharmonic modes (e.g., torsions, hydrogen bonds) require anharmonic Franck–Condon analysis.

19 Conclusion

The spectral hologram is not three spectra added together. It is the complete ternary state function in S-entropy space, recording both amplitude and phase of the molecular oscillatory system through three electronic states. From a single superimposed measurement, it yields six quantities inaccessible to any single-state

spectroscopy: the vibrational coupling matrix, Franck–Condon factors, Stokes shift decomposition, Huang–Rhys factors, Marcus reorganisation energy, and molecular point group symmetry.

The derivation proceeds from a single axiom (bounded phase space) through oscillatory necessity, S-entropy coordinates, emission-triggered temporal separation, and ternary projection completeness. No stored spectra, training data, or adjustable parameters enter the framework. The molecule is the instrument: its oscillators are the processors, its emission is the clock, its frequencies are the data.

Autocatalytic information gain provides SNR $\propto N^{0.69}$, a 2.7× enhancement over independent modalities. Categorical temporal resolution reaches $\delta t_{\mathrm{cat}} \sim 10^{-29}$ s through phase accumulation across the molecular oscillator ensemble, achieving sub-femtosecond state distinguishability without sub-femtosecond detectors.

Validation on CH_4^+ achieves 99.5% cross-prediction accuracy, zero mutual exclusion violation, and 0.983 trajectory fidelity. Validation on Rhodamine 6G reproduces Stokes shift, reorganisation energy, and Huang–Rhys factors within experimental uncertainty, and correctly identifies the C_2 point group from diffraction.

The hologram is empty. It stores nothing. It derives everything.

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